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A Simulation-Informed Grey-Box Degradation Model for Proton Exchange Membrane Fuel Cells

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Abstract

Proton exchange membrane fuel cells (PEMFCs) are essential functional parts for hybrid electrical vehicles. However, their long-term durability remains a critical engineering challenge, making effective degradation assessment essential for enabling health-aware energy management and extending fuel cell system lifetime. The existing degradation modeling approaches face a fundamental trade-off between physical interpretability and data efficiency. Therefore, hybrid grey-box models have become promising direction.

This thesis proposes a simulation-informed grey-box degradation assessment framework for PEMFC systems. Two interconnected problems are addressed: estimating internal operating condition indicators from externally measurable signals, and quantifying their contributions to stack voltage loss rate. Three internal variables are selected as degradation-relevant indicators: hydrogen excess ratio (HER), oxygen excess ratio (OER), and membrane water content, as they reflect reactant supply conditions and membrane hydration state.

A physics-based Simscape model of PEMFC system is first calibrated against experimental efficiency data using Bayesian optimization, and used as a high-fidelity data-generation tool. Supervised machine learning models are then trained using random forest regression to estimate HER, OER, and membrane water content from fuel-cell power-demand features. For HER and OER, a compact feature set based on instantaneous power and power slope achieved good predictive performance. By contrast, membrane water content showed slower dynamics and stronger load-history dependence, requiring a longer historical window to achieve reliable external generalization.

Finally, a grey-box degradation index is constructed by aggregating physics-informed severity terms derived from the estimated internal states and the external load profile. The degradation index is calibrated against experimentally measured voltage loss rates. The calibrated weights indicate that membrane hydration cycling, power cycling, oxygen-supply severity, and power amplitude are the dominant degradation-driving factors within the investigated operating profiles.

Overall, the proposed framework provides a physically interpretable and data-efficient approach to PEMFC degradation assessment, without requiring direct internal-state measurements or large-scale ageing datasets, providing a new perspective for degradation modelling.

Keywords

Proton exchange membrane fuel cell, Degradation modelling, Grey-box model, Machine learning, Random forest, Bayesian optimization, Hydrogen excess ratio, Oxygen excess ratio, Membrane water content, Heavy-duty transportation

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En simuleringsbaserad grå-box-nedbrytningsmodell för protonutbytesmembran-bränsleceller

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Sammanfattning

Protonutbytesmembran-bränsleceller (PEMFC) är viktiga funktionella komponenter i hybridelektriska fordon. Deras långsiktiga hållbarhet utgör dock en kritisk ingenjörsmässig utmaning, vilket gör effektiv nedbrytningsbedömning avgörande för hälsomedveten energihantering och förlängning av bränslecellssystemets livslängd. Befintliga modelleringsmetoder för nedbrytning står inför en grundläggande avvägning mellan fysikalisk tolkbarhet och dataeffektivitet. Därför har hybrida grå-box-modeller blivit en lovande inriktning.

Detta examensarbete föreslår ett simuleringsbaserat grå-box-ramverk för nedbrytningsbedömning av PEMFC-system. Två sammankopplade problem adresseras: att uppskatta interna driftsförhållandeindikatorer från externt mätbara signaler, samt att kvantifiera deras bidrag till stackspänningsförlusthastigheten. Tre interna variabler väljs som nedbrytningsrelevanta indikatorer: väteöverskottskvot (HER), syröverskottskvot (OER) och membranvatteninnehåll, eftersom dessa återspeglar reaktanttillförselförhållanden och membranhydrering.

En fysikbaserad Simscape-modell av PEMFC-systemet kalibreras först mot experimentella effektivitetsdata med hjälp av Bayesiansk optimering och används som ett högfidelitetsverktyg för datagenerering. Övervakade maskininlärningsmodeller tränas sedan med slumpskogsregression för att uppskatta HER, OER och membranvatteninnehåll utifrån bränslecellens effektefterfrågeegenskaper. För HER och OER uppnådde en kompakt funktionsuppsättning baserad på momentaneffekt och effektlutning god prediktiv prestanda. Membranvatteninnehållet visade däremot långsammare dynamik och starkare beroende av belastningshistorik, vilket krävde ett längre historiskt tidsfönster för tillförlitlig generalisering.

Slutligen konstrueras ett grå-box-nedbrytningsindex genom att aggregera fysikbaserade allvarlighetstermer härledda från de uppskattade interna tillstånden och den externa belastningsprofilen. Indexet kalibreras mot experimentellt uppmätta spänningsförlusthastigheter. De kalibrerade vikterna indikerar att membranhydreringscykling, effektcykling,

syrförsörjningsallvarlighet och effektamplitud är de dominerande nedbrytningsdrivande faktorerna inom de undersökta driftprofilerna.

Sammantaget erbjuder det föreslagna ramverket ett fysikaliskt tolkbart och dataeffektivt tillvägagångssätt för PEMFC-nedbrytningsbedömning, utan att kräva direkta interna tillståndsmätningar eller storskaliga åldringsdata, och bidrar därmed med ett nytt perspektiv på nedbrytningsmodellering.

Nyckelord

Protonutbytesmembran-bränslecell, Nedbrytningsmodellering, Grå-box-modell, Maskininlärning, Slumpskog, Bayesiansk optimering, Väteöverskottskvot, Syröverskottskvot, Membranvatteninnehåll, Tung transport

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Generative AI tools were used in this thesis to support the translation of the Swedish abstract, the development of part of programming and plotting code, and the correction of grammar and language errors.

Finally, I hope to carry the lessons learned into my future work and continue striving to become a responsible engineer who contributes with care, integrity, and perseverance.

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Dongyue Lei

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List of acronyms

BO	Bayesian optimization
BoP	balance of plant
CDF	cumulative distribution function
ECSA	electrochemical active surface area
EIS	electrochemical impedance spectroscopy
EMS	energy management system
GP	Gaussian process
GPR	Gaussian process regression
HER	hydrogen excess ratio
LHV	lower heating value
LSTM	long short-term memory
MAE	mean absolute error
MEA	membrane electrode assembly
ML	machine learning
MPC	model predictive control
MSE	mean squared error
OER	oxygen excess ratio
PEM	proton exchange membrane
PEMFC	proton exchange membrane fuel cell
RF	random forest
RH	relative humidity
RMSE	root mean square error
RUL	remaining useful life
SDG	Sustainable Development Goal
SOH	state of health
SVM	support vector machine

Chapter 1

Introduction

This chapter introduces the research background and motivation for the thesis. The specific research problems and questions are defined, followed by a description of the purpose, goals, methodology, and delimitations of the thesis. The structure of the thesis is also outlined in this chapter.

1.1 Background

With the increasing demand for low-carbon transportation, hybrid electric vehicles have become an important solution for reducing greenhouse gas emissions and improving fuel efficiency in the automotive industry[1, 2]. In a fuel-cell hybrid electric vehicle, the powertrain usually consists of a PEMFC system and an energy-storage device, such as a battery. The PEMFC system acts as an important onboard energy converter and is mainly responsible for supplying the average or sustained power demand [3]. As hybrid electrical vehicles approach commercial deployment, the long-term durability of PEMFC systems has become a critical engineering challenge. Degradation of the fuel cell stack, manifested as a gradual decline in output voltage over operating time, directly affects system lifetime and efficiency [4, 5, 6]. Effective degradation assessment is therefore essential for enabling health-aware energy management and extending the operational life of fuel cell systems.

Various PEMFC degradation modelling methods have been established, yet few of them can balance physical interpretability, dynamic applicability, and data efficiency [7]. This thesis is mainly focused on grey-box modeling and its application to PEMFC degradation assessment under dynamic operating conditions.

1.2 Problem

The central challenge addressed in this thesis is the construction of a PEMFC degradation assessment model that is both physically interpretable and data-efficient, without requiring direct measurement of internal stack variables or large-scale ageing datasets [6, 7]. This challenge involves two interconnected problems.

The first problem concerns the estimation of internal operating condition indicators from externally measurable signals. Since HER, OER, and membrane water content

cannot be directly obtained from standard sensors, a data-driven surrogate model must be constructed to infer these variables from the fuel cell power demand profile. The accuracy and generalization capability of such a model depend on the choice of regression method and input feature configuration, and it is not clear a priori which combination is most suitable for each internal state.

The second problem concerns the quantitative connection between these internal states and measurable degradation outcomes. Even if the internal operating condition indicators can be estimated, their individual contributions to the voltage loss rate of the stack under dynamic operating profiles remain unclear. A structured degradation index is needed to aggregate these indicators into a scalar severity score that can be calibrated against experimental degradation data.

These two problems lead to the following research questions investigated in this thesis:

- **RQ1:** How does the choice of regression model and input feature configuration affect the estimation accuracy of hydrogen excess ratio, oxygen excess ratio, and membrane water content under dynamic load profiles representative of heavy-duty fuel cell operation?
- **RQ2:** Since existing approaches rely on empirically assigned coefficients and have not been directly validated against measured voltage loss rates, how can the thresholds and weighting coefficients of a fuel cell degradation index be determined through data-driven calibration to quantify the contributions of hydrogen starvation, oxygen starvation, membrane hydration deviation, and power cycling to the voltage loss rate?

1.3 Purpose

This thesis aims to address two main objectives. From an academic perspective, it aims to establish a grey-box degradation assessment framework for PEMFC systems that retains physical interpretability while reducing the dependence on large-scale ageing datasets. From an engineering perspective, the resulting degradation index is intended to provide a physically meaningful health-related cost term that can support health-aware energy management strategies in fuel cell vehicles. The framework contributes to the broader goal of extending fuel cell system lifetime and reducing the lifecycle cost of hydrogen-powered heavy-duty vehicles.

1.4 Goals

The overall goal of this thesis is to develop and validate a grey-box degradation model for a PEMFC system operating under dynamic load profiles. This goal is divided into three sub-goals:

1. Develop and calibrate a physics-based Simscape model of the target PEMFC system using Bayesian optimization, and use the calibrated model as a high-fidelity data-generation tool for internal operating state variables.

2. Train and evaluate supervised machine learning models to estimate HER, OER, and membrane water content from externally available power-demand features, and systematically investigate the effect of regression model type and input feature configuration on estimation accuracy and generalization capability.

3. Construct and calibrate a degradation index by aggregating internal-state-based and load-profile-based severity terms, and validate its ability to reproduce experimentally measured voltage loss rates across diverse dynamic operating profiles.

1.5 Sustainability and Ethics

The development of durable and efficient fuel cell systems contributes to the decarbonization of the heavy-duty transport sector by supporting low-emission powertrain technologies and extending fuel-cell system lifetime [8, 2]. By improving degradation assessment and enabling health-aware operation, this work supports the extension of fuel cell system lifetime, which reduces material consumption, lowers replacement costs, and decreases the environmental impact associated with fuel cell manufacturing and disposal. These contributions are aligned with the United Nations Sustainable Development Goals, specifically SDG 7 (Affordable and Clean Energy) by reducing the lifecycle cost of hydrogen fuel cell systems and SDG 13 (Climate Action) by supporting the decarbonization of freight transport. Together, these contributions align with international efforts to achieve carbon neutrality in the transport sector.

From an ethical standpoint, this thesis adheres to the principles of academic transparency. The use of generative AI tools in the preparation of this work is disclosed in the acknowledgements section. All simulation data, model parameters, and experimental reference data used in this thesis are either publicly available or obtained from collaborating parties with appropriate permission.

1.6 Research Methodology

The methodology of this thesis integrates physics-based system modeling, machine learning, and degradation index formulation. A Simscape model of the PEMFC system is first calibrated against experimental efficiency data using Bayesian optimization, serving as a high-fidelity tool for generating internal operating state variables. Supervised regression models are then trained on simulation-generated data to estimate HER, OER,and membrane water content from externally available power-demand features, with random forest regression selected as the representative model based on its generalization performance. Finally, a grey-box degradation index is constructed by aggregating physics-informed severity terms derived from the estimated internal states and the external load profile, and calibrated against experimentally measured voltage loss rates. The components are systematically evaluated in terms of estimation accuracy, feature sensitivity, and degradation prediction performance under diverse dynamic operating profiles. Detailed methods and theoretical foundations are presented in Chapter 3.

1.7 Delimitations

This thesis is subject to the following delimitations. The physics-based Simscape model is calibrated using a single steady-state efficiency curve provided for the fuel cell system. Detailed internal stack parameters are not publicly available, and the model is therefore an approximation of the true system behaviour, with known residual mismatch in the low-power operating region.

All machine learning training data are generated from the calibrated simulation model. The estimated internal state variables, HER, OER, and membrane water content, are not independently validated against real sensor measurements, as these quantities are not directly accessible in practical systems.

The degradation index is calibrated and validated on a dataset of 53 operating profiles. This dataset may not include extreme conditions such as hydrogen starvation, low-temperature operation, or varied ageing stages. The generalizability of the calibrated weights to operating conditions outside this range may need further establishment.

1.8 Structure of the Thesis

The structure of this thesis is organised as follows. Chapter 2 provides the theoretical background on PEMFC system structure, internal operating state variables, and existing degradation modelling approaches, and identifies the research gap that motivates this work. Chapter 3 describes the overall methodological framework, including the physics-based system model, the machine learning surrogate, and the grey-box degradation index formulation. Chapter 4 presents the results of the parameter identification, machine learning model evaluation, and degradation index validation, together with analysis and discussion. Chapter 5 summarises the conclusions of the thesis, identifies limitations, and outlines directions for future work.

Chapter 2

Foundations of PEMFC Systems and Degradation Modelling

This chapter provides basic background information about PEM fuel system and structure, as well as three internal state variables that are strongly affected by the operating conditions of a PEM fuel cell: the hydrogen excess ratio (HER), the oxygen excess ratio (OER), and the membrane water content(λ_w). This chapter also reviews mechanism and related work on PEM fuel cell degradation modelling and health-state prediction

2.1 PEM fuel cell system

The PEMFC system can be regarded as an integrated energy conversion system composed of three main subsystems: the hydrogen supply path, the air supply path, and the fuel cell stack. As shown in Figure 2.1, hydrogen is supplied from the hydrogen tank to the anode side of the stack after pressure regulation and humidification, while air is delivered to the cathode side through compression and humidification. Inside the stack, the anode, proton exchange membrane, and cathode are arranged as the core electrochemical conversion unit, where the chemical energy of hydrogen is converted into electrical energy supplied to the external load. Overall, the PEMFC system is formed by coupling the gas supply subsystems with the electrochemical stack and the external electrical circuit, providing a complete pathway from reactant supply to electrical power output.

2.1.1 Fuel cell stack

The fuel cell stack is the core electrochemical component of a PEM fuel cell system. It converts the chemical energy stored in hydrogen into electrical energy through electrochemical reactions. A stack consists of multiple single cells connected in series, so that the total stack voltage is obtained by multiplying the single-cell voltage by the number of cells. In each cell, hydrogen is supplied to the anode side, while oxygen is supplied to the cathode side. At the anode, hydrogen molecules are split into protons and electrons. The protons are transported through the proton exchange membrane, while the electrons flow through the external circuit and provide electrical power to the load. At the

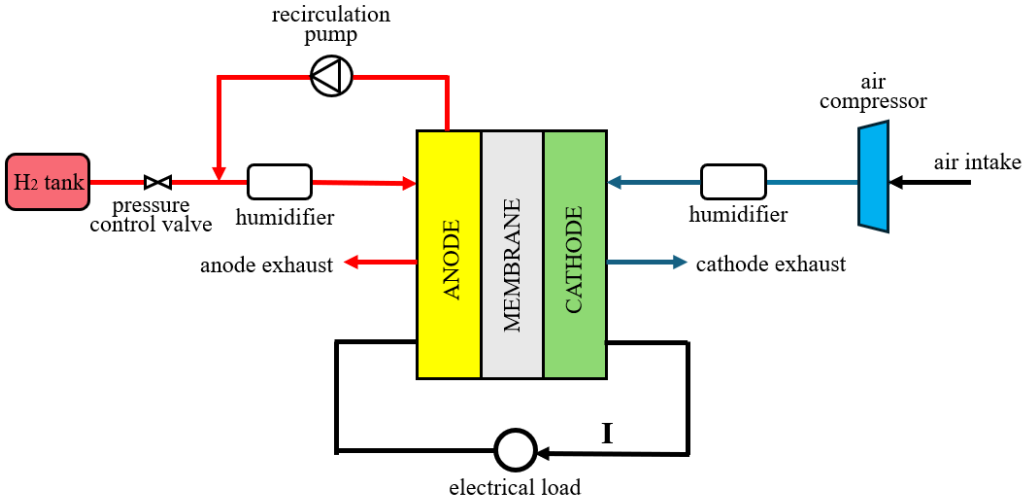


Figure 2.1: Schematic diagram of a PEM fuel cell system

cathode, oxygen reacts with the transported protons and electrons to form water.

The main electrochemical reactions inside a PEM fuel cell can be written as

$$H_2 \rightarrow 2H^+ + 2e^- \tag{2.1}$$

$$\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O \tag{2.2}$$

$$H_2 + \frac{1}{2}O_2 \rightarrow H_2O \tag{2.3}$$

where the first equation represents the anode reaction, the second equation represents the cathode reaction, and the third equation represents the overall reaction of the fuel cell.

2.1.2 Hydrogen supply path

The hydrogen supply path provides the anode side of the fuel cell stack with sufficient hydrogen under controlled pressure and humidity conditions. In a PEM fuel cell system, hydrogen is usually stored in a high-pressure tank and then delivered to the anode inlet through a pressure regulation stage. The pressure regulator reduces the tank pressure to the required operating pressure of the stack, while the flow control components adjust the supplied hydrogen flow rate according to the demanded stack current.

2.1.3 Air supply path

The air supply path provides oxygen to the cathode side of the stack. In practical PEM fuel cell systems, ambient air is usually compressed before entering the cathode channels, because increasing the cathode pressure can improve oxygen availability and enhance

stack performance. The air supply subsystem typically includes an air filter, compressor, humidifier, valves, and cathode flow channels. Among these components, the compressor is one of the most important balance-of-plant components, as it directly affects both the oxygen supply condition and the parasitic power consumption of the system.

2.1.4 Internal operating state variables

Besides external outputs such as stack power and efficiency, internal operating state variables are needed to describe the dynamic condition of a PEM fuel cell stack. In this thesis, the hydrogen excess ratio (HER), oxygen excess ratio (OER), and membrane water content λ_w are selected as the main internal state variables. These variables represent reactant supply and membrane hydration conditions, and therefore provide a physically meaningful basis for analysing degradation-related operation under dynamic load profiles.

HER and OER describe whether the supplied hydrogen and oxygen flow rates are sufficient compared with the theoretical reactant consumption required by the electrochemical reaction. Since the reactant consumption rate is directly determined by the stack current, these two ratios can be expressed using Faraday's law as

$$HER = \frac{\dot{n}_{\mathrm{H_2,in}}}{\dot{n}_{\mathrm{H_2,cons}}} = \frac{2F\dot{n}_{\mathrm{H_2,in}}}{N_{\mathrm{cell}} I_{\mathrm{stack}}}, \tag{2.4}$$

$$OER = \frac{\dot{n}_{\mathrm{O_2,in}}}{\dot{n}_{\mathrm{O_2,cons}}} = \frac{4F\dot{n}_{\mathrm{O_2,in}}}{N_{\mathrm{cell}} I_{\mathrm{stack}}}, \tag{2.5}$$

where $\dot{n}_{\mathrm{H_2,in}}$ and $\dot{n}_{\mathrm{O_2,in}}$ are the inlet molar flow rates of hydrogen and oxygen, respectively, N_{cell} is the number of cells in the stack, I_{stack} is the stack current, and F is Faraday's constant.

An excess ratio equal to one means that the supplied reactant flow is exactly equal to the stoichiometric requirement. In practical operation, HER and OER are maintained above one to provide a safety margin against reactant starvation. A low HER indicates possible hydrogen starvation on the anode side, while a low OER indicates possible oxygen starvation on the cathode side. Both conditions can increase voltage losses and contribute to accelerated stack degradation.

Under dynamic load profiles, HER and OER may fluctuate because the stack current changes faster than the gas supply subsystems can respond. During rapid load increases, the temporary mismatch between reactant consumption and supply can reduce the excess ratios. Therefore, HER and OER are useful indicators for evaluating transient starvation risk and degradation-prone operating conditions.

Another important internal process in the fuel cell stack is membrane water transport. The membrane water content, denoted by λ_w, describes the hydration level of the proton exchange membrane. It is usually defined as the number of water molecules per sulfonic acid site in the membrane. The membrane water balance is affected by several physical mechanisms, including water generation at the cathode, electro-osmotic drag from anode to cathode, back diffusion driven by water concentration gradients, and water removal by the gas flow. A simplified dynamic representation of the membrane water balance can be written as

$$\frac{d\lambda_w}{dt} = f_{\text{gen}} - f_{\text{drag}} + f_{\text{diff}} - f_{\text{remove}} \tag{2.6}$$

where f_{gen} represents water generation, f_{drag} represents electro-osmotic drag, f_{diff} represents back diffusion, and f_{remove} represents water removal through the gas channels. Membrane water management is essential for maintaining stable stack performance. Insufficient water content causes membrane dehydration and increases ohmic resistance, while excessive water accumulation may lead to flooding and stronger mass transport limitation.

2.2 Fuel cell degradation mechanism

PEM fuel cell degradation is a coupled ageing process involving electrochemical, chemical, mechanical, thermal, and mass-transport effects. At the system level, it is usually observed as a gradual decrease in stack voltage under the same operating condition. This voltage loss can be caused by degradation of several components, including the catalyst layer, proton exchange membrane, gas diffusion layer, and bipolar plates [4, 5, 9]. Since these mechanisms are strongly affected by operating conditions, PEMFC degradation cannot be fully described by operating time alone. Instead, the load profile, reactant supply condition, water-management state, and transient operation all need to be considered [10, 8, 2].

One important degradation mechanism is catalyst-layer degradation. During long-term operation, the electrochemical active surface area of the platinum catalyst may decrease because of platinum dissolution, agglomeration, Ostwald ripening, and detachment from the carbon support. This reduces the available reaction area and increases activation losses. In addition, carbon corrosion can damage the catalyst support, weaken the electronic and ionic transport network, and further accelerate the loss of catalytic activity [4, 9, 11]. These effects are particularly relevant during high-potential operation, load cycling, start-stop events, and reactant starvation [5, 2].

Membrane degradation is another major source of performance loss. The proton exchange membrane must remain sufficiently hydrated to maintain high proton conductivity. Insufficient membrane water content causes membrane dehydration, which increases ohmic resistance and reduces cell voltage. In contrast, excessive water accumulation may lead to flooding, blocking gas transport pathways and increasing concentration losses. In addition, repeated hydration and dehydration cycles can produce mechanical stress in the membrane, leading to crack formation, pinholes, gas crossover, and long-term membrane failure [4, 6, 11]. Therefore, membrane water content is closely related to both instantaneous performance and long-term durability.

Reactant starvation is also strongly related to PEMFC degradation. When the supplied hydrogen or oxygen flow rate is insufficient compared with the electrochemical consumption rate, local fuel or oxidant starvation may occur. Hydrogen starvation can cause local cell reversal and high electrode potentials, which may accelerate carbon corrosion and irreversible performance loss on the anode side [12]. Oxygen starvation can increase cathode overpotential, enhance mass-transport losses, and contribute to catalyst-

layer degradation. Therefore, the hydrogen excess ratio and oxygen excess ratio are useful indicators of the reactant-supply margins on the anode and cathode sides, respectively.

Dynamic power operation can further accelerate degradation because it repeatedly changes current density, reactant consumption, water generation, and heat production. In vehicle applications, load cycling is one of the main differences between fuel cells used in transportation and those used in stationary applications, and it is closely related to water management and gas-transport problems [8]. Rapid load increases may cause a temporary mismatch between reactant demand and gas-supply response, while large-amplitude power cycling can disturb membrane hydration and pressure conditions. As a result, degradation under dynamic operation should be analysed using both instantaneous operating conditions and load-history characteristics [10, 2].

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2.3 Degradation modelling of PEMFC systems

Existing PEMFC degradation modelling methods can generally be classified into three major categories: physics-based models, data-driven models, and hybrid models [7]. Each category has different advantages and limitations in terms of physical interpretability, data requirement, computational complexity, and applicability to dynamic operating conditions. The following subsections review these three categories and discuss how they motivate the grey-box degradation assessment framework adopted in this thesis.

2.3.1 Physics-based degradation models

Physics-based degradation models describe PEMFC behaviour using electrochemical, thermodynamic, gas-transport, and polarization-curve-related equations. Degradation is usually represented by ageing parameters introduced into physically meaningful quantities, such as electrochemical active surface area (ECSA), internal resistance, open-circuit voltage, limiting current, activation losses, ohmic losses, or mass transport losses. The main advantage of these models is their physical interpretability, since parameter variations can be related to degradation mechanisms inside the fuel cell stack. In addition, they usually require less training data than purely data-driven models [7].

Several representative approaches have been reported. Zhang and Pisu [13] used ECSA as an ageing parameter and applied an unscented Kalman filter to estimate its degradation. Zhang et al. [14] and Bressel et al. [15] introduced equivalent ageing variables into polarization-curve parameters, such as internal resistance, open-circuit voltage, and limiting current, and used filtering methods for health estimation and prognosis. These models are relatively simple and suitable for online estimation, but they compress multiple degradation mechanisms into a limited number of parameters.

More detailed models have also been developed. Zhou et al. [16] introduced ageing coefficients associated with ohmic loss, reaction activity loss, and mass transport loss. Jouin et al. [17] further considered multiple ageing-related parameters, including charge transfer coefficients, hydrogen crossover current, ionic resistance, contact resistance, water/gas accumulation effects, and cathode limiting current. These models improve

physical resolution, but they require more detailed parameter information and still rely largely on quasi-static polarization behaviour.

Overall, physics-based degradation models provide valuable physical insight and relatively good data efficiency. However, many of them are static or quasi-static, require detailed internal stack parameters, and may not fully capture transient phenomena under dynamic load profiles. These limitations motivate the use of data-driven or hybrid modelling strategies for degradation assessment under realistic operating conditions.

2.3.2 Data-driven degradation models

Data-driven degradation models learn the relationship between measured inputs and degradation-related outputs directly from data. They do not require detailed knowledge of internal electrochemical parameters or explicit degradation laws, which is useful when the internal structure of a commercial PEMFC stack is unknown. However, because they lack explicit physical structure, they often behave as black-box models with limited interpretability and weak extrapolation capability outside the training range [7].

Neural-network-based models have been widely used for PEMFC performance and degradation prediction. Napoli et al. [18] used a feedforward neural network to predict stack voltage and cathode temperature, while Chen et al. [19, 20, 21] proposed several neural-network-based degradation prediction and ageing prognosis models. These studies show that neural networks can capture nonlinear relationships between operating variables and degradation-related performance loss, but their reliability depends strongly on the amount and representativeness of ageing data.

For time-dependent degradation behaviour, recurrent and reservoir-based models have also been applied. Liu et al. [22] used an LSTM network for PEMFC prognostics, and Ma et al. [23] applied deep learning to degradation prediction. Morando et al. [24], Hua et al. [25], and Vichard et al. [26] used echo state networks to forecast stack voltage evolution and remaining useful life. These methods can model temporal degradation trends, but they still require training data that sufficiently cover the relevant operating and environmental conditions.

Electrochemical impedance spectroscopy (EIS) and equivalent circuit models have also been used for PEMFC diagnosis and state-of-health monitoring. Lee et al. [27] used an EIS-based equivalent impedance model to estimate PEMFC lifespan, while Fouquet et al. [28] proposed SOH monitoring based on AC impedance measurements. EIS can reveal information related to ohmic resistance, charge transfer, diffusion, drying, and flooding [29, 30]. However, it is sensitive to operating conditions and usually requires a fixed operating point, which limits its use in dynamic vehicle applications.

Overall, data-driven models are effective when sufficient representative data are available, but they require large ageing datasets, generalize poorly to unseen conditions, and provide limited physical insight. For degradation-aware operation, this is a key limitation because the model should not only predict performance loss, but also indicate whether the risk is related to gas starvation, water management, high-power operation, or power cycling.

2.3.3 Hybrid degradation models

Hybrid degradation models aim to combine physics-based structure with data-driven flexibility. In principle, the physics-based part provides interpretability and generalization, while the data-driven part learns unknown nonlinear relationships or compensates for modelling errors. This is attractive for PEMFC degradation modelling because complete degradation laws are difficult to obtain, whereas purely black-box models usually require large ageing datasets and provide limited physical insight [7].

Several hybrid or semi-hybrid PEMFC models have been reported. Lan and Strunz [31] proposed a multi-physics transient model using equivalent circuits to represent coupled electrical, pneumatic, and thermal behaviour. Although it is not a pure degradation-law model, it provides a useful framework for describing transient fuel-cell dynamics that are relevant to degradation-related states. Lechartier et al. [32] developed a PEMFC behavioural model for prognostics, combining a static physics-based part with a dynamic electrical-equivalent part, with parameters updated through polarization curves and EIS measurements. Zhong et al. [33] proposed a hybrid multi-variable PEMFC model that combines least-squares support vector machines with a pressure-incremental physics-based model to account for current, temperature, and pressure effects.

Overall, hybrid models are promising because they can potentially retain physical transparency while improving adaptability to nonlinear and dynamic behaviour. However, they also require a careful balance between physical structure and data-driven components. If the physical part is too simplified, important transient effects may be missed; if the data-driven part dominates, the model may lose interpretability and become highly data-dependent. Moreover, the number of published PEMFC hybrid degradation models remains limited, and dynamic, interpretable, and data-efficient hybrid degradation assessment is still an open research direction.

2.3.4 Research gap and motivation of thesis

The literature review above indicates that PEMFC degradation modelling still faces a trade-off between physical interpretability, dynamic applicability, and data requirement. Physics-based models provide interpretable ageing parameters, but many of them are quasi-static and require detailed internal stack information. Data-driven models can capture nonlinear dynamic behaviour, but they usually require large representative ageing datasets and provide limited physical insight. Hybrid models are therefore considered a promising direction because they can combine physical structure with data-driven flexibility, although the number of published PEMFC hybrid degradation models remains limited [7, 32, 33].

A closely related health-aware MPC study has introduced hydrogen excess ratio (HER), oxygen excess ratio (OER), and membrane water content as degradation-related penalty terms in the control objective [3]. It provides a useful grey-box perspective for degradation assessment: degradation-prone operation can be represented through physically meaningful internal state variables, rather than relying solely on detailed material-level ageing models or purely empirical degradation labels. These three variables represent the main operating conditions associated with PEMFC degradation. HER and

OER describe the reactant-supply margin and are directly related to hydrogen and oxygen starvation risk, while membrane water content reflects water-management conditions, including membrane drying, flooding tendency, hydration cycling, and the resulting ohmic behaviour [7, 6, 3]. Although they do not directly measure microscopic material ageing, they provide physically interpretable indicators of degradation-prone operation.

However, the work in [3] is mainly control-oriented. The resulting health cost is primarily used as an MPC penalty rather than being developed as an independent degradation assessment framework, and its quantitative relationship with actual degradation indicators, such as measured voltage decay or long-term ageing data, is not directly validated. The model mainly relies on instantaneous power information and does not explicitly consider historical load effects. In addition, the influence of different regression methods and input feature sets on internal-state prediction accuracy is not systematically investigated.

Therefore, the contribution of this thesis is twofold. First, it proposes a grey-box degradation assessment approach that is both physically interpretable and data-efficient. HER, OER, and membrane water content are retained as intermediate health-related states, while machine learning is used to estimate them from measurable dynamic load-profile features. Second, the framework further connects these internal states to degradation-relevant indicators rather than using them only as control penalty terms. Historical load-window descriptors are also introduced into the machine-learning model to examine whether load-history effects improve internal-state prediction. In this way, a physically interpretable and data-efficient degradation-oriented assessment model is established, which links measurable power-demand information to physical degradation quantity without requiring large-scale ageing datasets.

Chapter 3

Methodology for Simulation-Informed PEMFC Degradation Assessment

The purpose of this chapter is to describe the three-stage methodological framework developed in this thesis to construct a simulation-informed grey-box degradation model for PEM fuel cell systems. Section 3.1 provides an overview of the overall framework and explains how the three stages are connected. Section 3.2 describes the physics-based PEMFC system model implemented in Simscape and the Bayesian optimization procedure used to calibrate the model against available experimental efficiency data. Section 3.3 details the supervised machine learning models trained to estimate the internal operating condition indicators from externally available power-demand features. Section 3.4 presents the grey-box degradation index formulation and the calibration approach used to connect the degradation score to experimentally measured voltage loss rates.

3.1 Overall methodological framework

The overall methodology of this thesis is designed to establish a modelling framework that connects fuel cell operating conditions, internal state variables, and degradation cost, as illustrated in Figure 3.1.

First, a PEM fuel cell system model is applied in Simscape to represent the electrochemical and thermal-fluid behavior of the fuel cell system. Since detailed manufacturer parameters of the investigated fuel cell stack are not publicly available, the physical model is calibrated using available experimental fuel cell data.

Secondly, the physical model is used as a data-generation tool. Different power demand profiles are applied to the Simscape model, and the corresponding internal state variables are extracted, including hydrogen excess ratio (HER), oxygen excess ratio (OER), and membrane water content. These variables are selected because they reflect important gas supply and water management conditions inside the PEM fuel cell, and are closely related to degradation mechanisms such as reactant starvation and membrane hydration imbalance.

A machine learning model is then trained to approximate the mapping from load data

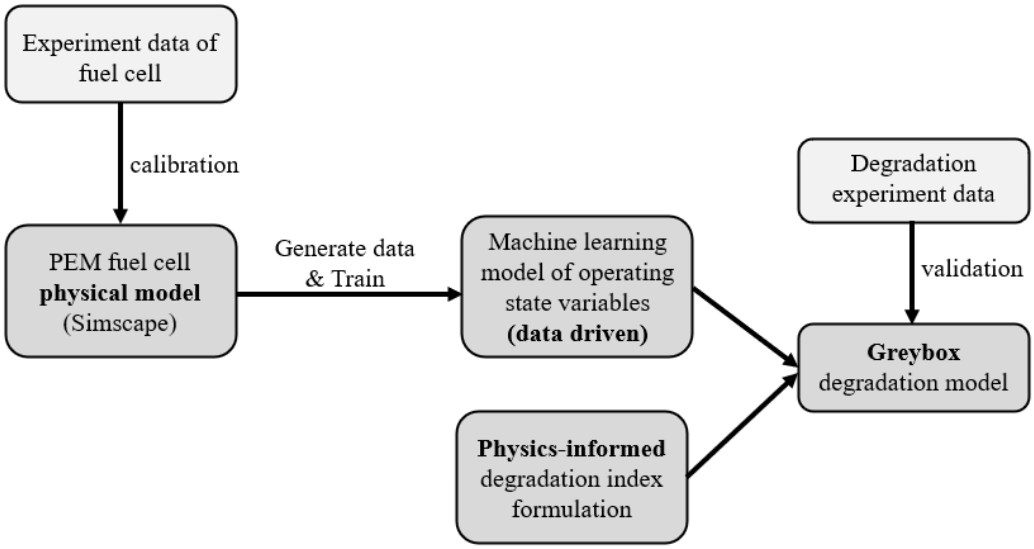


Figure 3.1: Overall methodological framework

to these internal state variables. This step is necessary because HER, OER, and membrane water content are not directly available from standard real-time sensors in practical fuel cell operation. It should be noted that the machine learning model is not employed as a black-box lifetime predictor. Instead, it is used as a data-driven surrogate estimator to infer physically interpretable internal operating states from accessible signal.

Finally, the internal state estimation is combined with a physics-informed degradation index, creating an estimation of voltage loss rate. The model is called grey-box because it integrates a data-driven component with a transparent physics-informed component. The resulting degradation model is then validated using available degradation experiment data. Through this validation, the thesis evaluates whether the proposed framework can reproduce the observed degradation value under different operating conditions.

3.2 Fuel Cell Physical Model

3.2.1 System modeling and Simscape implementation

The PEM fuel cell system implemented in the Simscape physical modeling environment. As shown in Figure **??**,the anode side receives hydrogen from the supply source, which is further regulated through the recirculation subsystem and humidified before entering the anode gas channels. On the cathode side, oxygen is supplied through the cathode humidifier and then delivered to the cathode gas channels. The anode and cathode gas channel states are used as inputs to the MEA block, where the electrochemical reaction, gas consumption, water transport, voltage generation, and heat production are calculated.

The internal variables are exported for analysis and degradation-related modeling,

including hydrogen excess ratio (HER), oxygen excess ratio (OER), membrane water content, stack voltage, current density, efficiency, theoretical cell voltage, and individual voltage losses. The balance-of-plant power consumption is also considered in the system-level efficiency calculation.

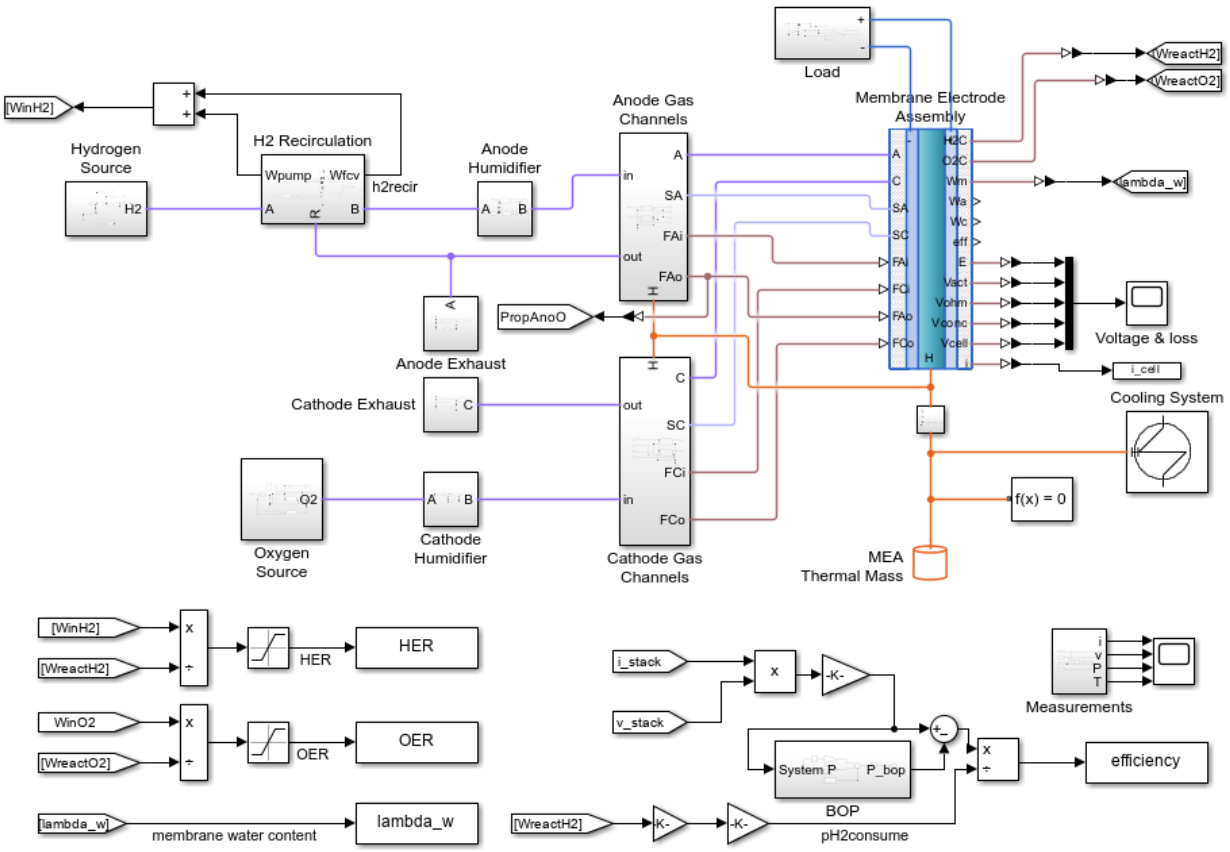


Figure 3.2: Simscape model of PEM fuel cell system

Fuel cell stack

The MEA is the core component of the model, representing the chemical reaction and power generartion in the fuel cell stack. It calculates the single-cell reversible voltage and then subtracts voltage losses. Meanwhile, the model computes hydrogen and oxygen consumption rates according to the stack current, and estimates membrane water content based on the local water activity at the anode and cathode catalyst layers. The generated heat is coupled to the thermal mass and cooling system, allowing the model to represent the interaction between electrochemical performance and thermal behavior.

The ideal reversible voltage of the cell is described by the Nernst equation, which relates the open-circuit voltage to temperature and reactant partial pressures:

$$E_{\text{Nernst}} = E^0 + \frac{R_u T}{2F} \ln\left(\frac{p_{\text{H}_2}\sqrt{p_{\text{O}_2}}}{p_{\text{H}_2\text{O}}}\right) \tag{3.1}$$

where E^0 is the standard reversible voltage, R_u is the universal gas constant, T is the stack temperature, F is Faraday's constant, and p_{H_2}, p_{O_2}, and $p_{\text{H}_2\text{O}}$ are the partial pressures of hydrogen, oxygen, and water vapour, respectively. This equation indicates that the theoretical cell voltage is not constant, but depends on the thermodynamic state of the reactants and products inside the stack.

In practical operation, the actual cell voltage is lower than the reversible voltage because of several irreversible loss mechanisms. The cell voltage can therefore be expressed as

$$V_{\text{cell}} = E_{\text{Nernst}} - V_{\text{act}} - V_{\text{ohm}} - V_{\text{conc}} \tag{3.2}$$

where V_{act} is the activation loss, V_{ohm} is the ohmic loss, and V_{conc} is the concentration loss. These losses describe different physical limitations during the electrochemical energy conversion process.

The activation loss represents the voltage loss required to overcome the electrochemical reaction barrier at the electrodes. It is especially important at low current density, where the reaction kinetics dominate the polarization behaviour. The activation loss is described by a Tafel-type relationship:

$$V_{\text{act}} = \frac{R_u T}{2\alpha F} \ln\left(\frac{i_{\text{cell}}}{i_0}\right) \tag{3.3}$$

where i_{cell} is the current density, i_0 is the exchange current density, and α is the charge transfer coefficient. A lower exchange current density or a higher operating current density leads to a larger activation overpotential. Therefore, this term mainly affects the low-current and medium-current regions of the polarization curve.

The ohmic loss is caused by the resistance to proton transport through the membrane and electron transport through the electrodes, bipolar plates, and external connections. It is commonly written as

$$V_{\text{ohm}} = i_{\text{cell}} R_{\text{ohm}} \tag{3.4}$$

where R_{ohm} is the area-specific ohmic resistance. For the membrane contribution, the resistance can be expressed as

$$R_{\text{ohm}} = \frac{t_{\text{mem}}}{\sigma_{\text{mem}}} \tag{3.5}$$

where t_{mem} is the membrane thickness and σ_{mem} is the membrane proton conductivity. The membrane conductivity is strongly influenced by temperature and membrane hydration. A well-hydrated membrane has higher proton conductivity and lower ohmic loss, whereas membrane dehydration increases the internal resistance and reduces the cell voltage.

The concentration loss describes the voltage reduction caused by mass transport

limitations. When the current density becomes high, the reactants may not be transported to the catalyst layers sufficiently fast, leading to a decrease in local reactant concentration. This effect can be represented as

$$V_{\text{conc}} = -\frac{R_u T}{2F} \ln\left(1 - \frac{i_{\text{cell}}}{i_L}\right) \tag{3.6}$$

where i_L is the limiting current density. As the current density approaches the limiting current density, the concentration loss increases rapidly. Therefore, this loss is particularly important in the high-power region of the polarization curve.

The total stack voltage can be obtained by multiplying the single-cell voltage by the number of cells:

$$V_{\text{stack}} = N_{\text{cell}} V_{\text{cell}} \tag{3.7}$$

where N_{cell} is the number of cells connected in series. The stack electrical power can be calculated as

$$P_{\text{stack}} = V_{\text{stack}} I_{\text{stack}} \tag{3.8}$$

where I_{stack} is the stack current. These equations show that the stack output power is directly determined by the electrochemical voltage behaviour and the load current.

The reactant consumption of gas can be derived based on Faraday's law. The molar consumption rates of hydrogen and oxygen are proportional to the stack current:

$$\dot{n}_{\text{H}_2,\text{cons}} = \frac{N_{\text{cell}} I_{\text{stack}}}{2F} \tag{3.9}$$

$$\dot{n}_{\text{O}_2,\text{cons}} = \frac{N_{\text{cell}} I_{\text{stack}}}{4F} \tag{3.10}$$

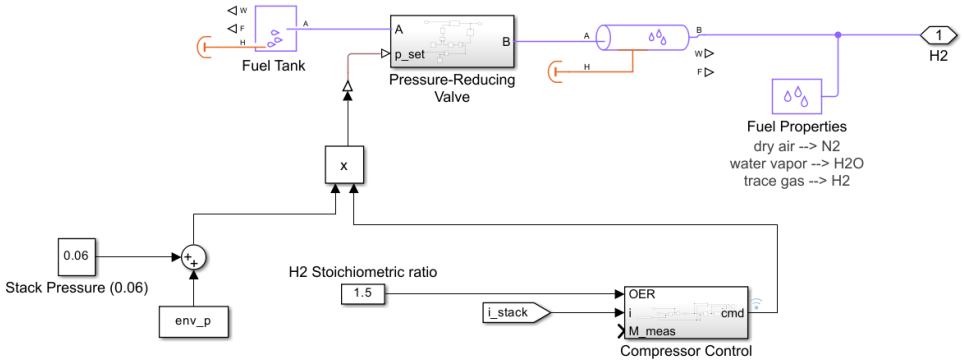
Similarly, the water generation rate is

$$\dot{n}_{\text{H}_2\text{O},\text{gen}} = \frac{N_{\text{cell}} I_{\text{stack}}}{2F} \tag{3.11}$$

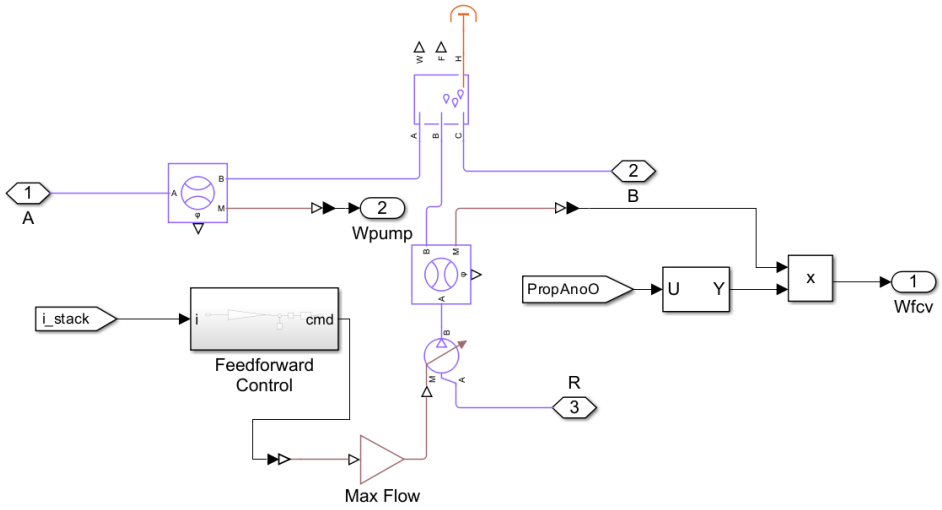
These expressions describe the coupling between the electrical load and the chemical reactant consumption, and are implemented in the membrane electrode assembly component.

Hydrogen supply and recirculation

The anode gas supply subsystem consists of two parts: the hydrogen supply path and the recirculation loop, as shown in Figures 3.3a and 3.3b. On the supply side, hydrogen is drawn from a high-pressure fuel tank and regulated by a pressure-reducing valve. The valve set pressure is determined by a target hydrogen stoichiometric ratio of 1.5, which is converted to a pressure command to maintain the desired stoichiometric ratio at the anode inlet. The recirculation loop returns unconsumed hydrogen via a pump driven by a feedforward controller based on the stack current i_{stack}, with a maximum flow limiter applied to prevent unrealistic transient flow rates.



(a) Hydrogen supply path.



(b) Hydrogen recirculation loop.

Figure 3.3: Simscape model of the anode gas supply and recirculation subsystem.

Oxygen supply

The cathode gas supply subsystem compresses ambient air and delivers it to the cathode gas channels, as shown in Figure 3.4. A compressor is driven by a control block that receives the target oxygen stoichiometric ratio (set to 1.5), the stack current i_{stack}, and the measured air mass flow rate as inputs. The required air flow is computed from the stoichiometric ratio and the stack current via Faraday's law, and the compressor speed is adjusted accordingly subject to a maximum speed constraint. A buffer volume between the compressor outlet and the cathode channels attenuates pressure transients under rapid load changes.

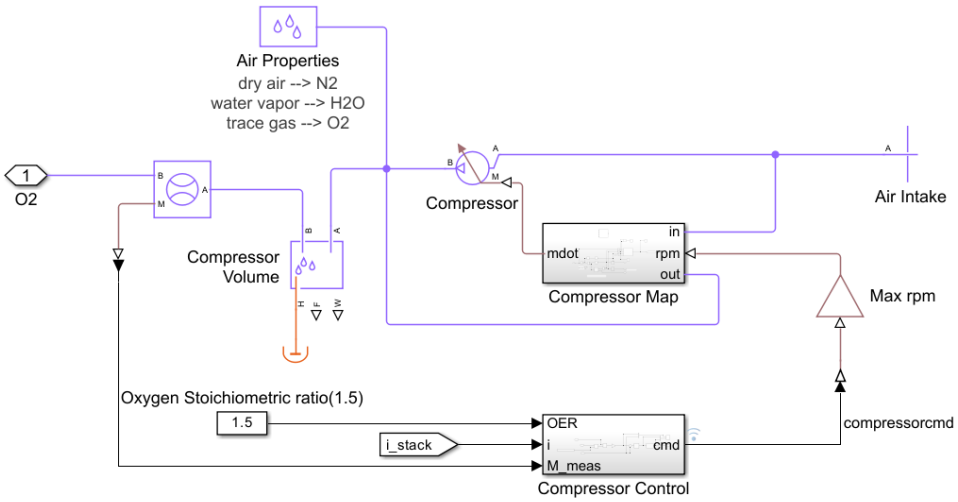


Figure 3.4: Simscape model of the cathode gas supply subsystem.

3.2.2 Parameter calibration of physical model

The fuel cell system considered in this work is a product from a heavy-duty truck manufacture. Due to proprietary constraints, its detailed parameters are not publicly available. Therefore, a parameter identification approach is adopted to calibrate the Simscape model, ensuring that the simulated behavior matches available reference data.

The optimization parameters, detailed in Table 3.1, are chosen to capture the dominant physical mechanisms of the fuel cell system. These parameters are categorized by their specific influence on performance: electrochemical (i_0, i_L, α) and membrane (t_{mem}) properties define voltage losses; geometric ($A_{\text{cell}}, N_{\text{cell}}$) factors determine system scaling; while mass transport and operating conditions ($D_{\text{GDL}}, \lambda_{\text{O}_2}, \text{RH}_{\text{cat}}$) govern gas supply and water management. Since the available reference data represent system-level efficiency, an empirical BoP correction is introduced to approximate the auxiliary power consumption of the complete fuel cell system. The BoP power is described as a quadratic function of the output power:

$$P_{\text{BoP}}(P) = a_0 + a_1 P + a_2 P^2 \tag{3.12}$$

where a_0, a_1, and a_2 are empirical correction coefficients. This term is used to represent the lumped influence of auxiliary components, such as air supply, cooling, hydrogen circulation, pumps, and power electronics.

Table 3.1: Parameters selected for Bayesian optimization and their bounds

Parameter	**Symbol**	**Initial Value**	**Range**
Exchange current density	i_0	1.52×10^{-5}	$[1 \times 10^{-5},\ 6 \times 10^{-5}]$
Limiting current density	i_L	1.80	[1.0, 2.0]
Membrane thickness (μm)	t_{mem}	118.56	[100, 160]
Effective cell area (cm^2)	A_{cell}	600.72	[500, 700]
Number of cells	N_{cell}	826	[780, 840]
Charge transfer coefficient	α	0.443	[0.40, 0.58]
Gas diffusion coefficient (cm^2/s)	D_{GDL}	0.038	[0.02, 0.05]
Oxygen stoichiometric ratio	λ_{O_2}	1.5	[1.0, 1.85]
Cathode relative humidity	RH_{cat}	0.716	[0.70, 0.80]
BoP constant term (kW)	a_0	2	[0, 5]
BoP linear coefficient	a_1	0.02	[0.02, 0.2]
BoP quadratic coefficient (1/kW)	a_2	5×10^{-5}	$[2 \times 10^{-5},\ 1.2 \times 10^{-4}]$

The purpose of this optimization loop is to minimizes the discrepancy between the simulated PEMFC system efficiency and the reference efficiency–power curve from the truck manufacturer. The objective function is defined as the weighted mean squared error between the simulated and reference efficiencies:

$$J(\mathbf{x}) = \frac{\sum_{k=1}^{N} w_k \left[\eta_{\text{sim}}(P_k; \mathbf{x}) - \eta_{\text{ref}}(P_k)\right]^2}{\sum_{k=1}^{N} w_k}, \tag{3.13}$$

where P_k is the k-th reference operating power point, $\eta_{\text{sim}}(P_k; \mathbf{x})$ is the simulated efficiency obtained from the Simscape model using parameter vector $\mathbf{x}$, $\eta_{\text{ref}}(P_k)$ is the corresponding reference efficiency, and w_k is the weighting factor assigned to that operating point.

Bayesian optimization is adopted because each objective evaluation requires a computationally expensive Simscape simulation, and the resulting objective function is nonlinear, non-convex, and unavailable in closed form. Compared with exhaustive search or gradient-based methods, Bayesian optimization can efficiently identify promising parameter regions with a limited number of function evaluations. The Bayesian optimization strategy used in this work follows the workflow shown in Fig. 3.5.

First, the search space of the optimization problem is defined by the lower and upper bounds of the selected stack parameters as shown in Table 3.1.

At each iteration, the Bayesian optimization engine uses all previously evaluated parameter sets and their corresponding objective values. After n evaluations, this evaluated dataset is denoted as

$$D_n = \{(\mathbf{x}_i, J_i)\}_{i=1}^{n}, \tag{3.14}$$

where $\mathbf{x}_i$ is the i-th evaluated parameter vector and J_i is the corresponding objective

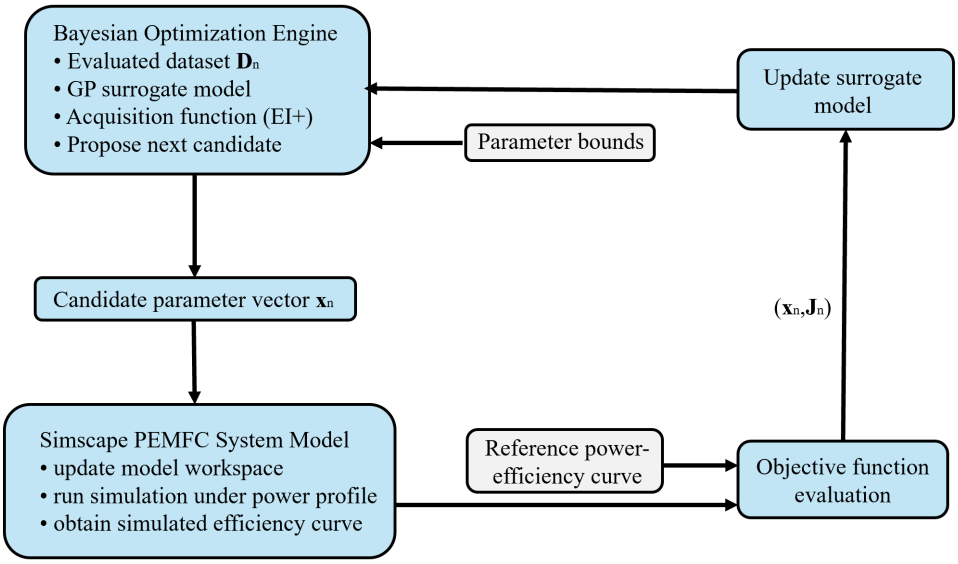


Figure 3.5: Workflow of the Bayesian-optimization process

value obtained from the Simscape model evaluation. Based on the evaluated dataset D_n, Bayesian optimization constructs a Gaussian-process surrogate model to approximate the unknown objective function. The objective function is modeled as

$$J(\mathbf{x}) \sim \mathrm{GP}\left(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')\right), \tag{3.15}$$

where $m(\mathbf{x})$ is the mean function and $k(\mathbf{x}, \mathbf{x}')$ is the covariance kernel function. After conditioning on the evaluated dataset D_n, the Gaussian-process surrogate gives a posterior predictive distribution for any candidate parameter vector:

$$J(\mathbf{x})|D_n \sim \mathrm{N}\left(\mu_n(\mathbf{x}), \sigma_n^2(\mathbf{x})\right). \tag{3.16}$$

In which $\mu_n(\mathbf{x})$ represents the predicted objective value, while $\sigma_n(\mathbf{x})$ represents the uncertainty of this prediction. Therefore, the surrogate model provides not only an estimate of the fitting cost, but also a measure of how uncertain the optimizer is about this estimate.

The next candidate parameter vector is then selected by the acquisition function. In this work, the expected-improvement-plus acquisition function is used. The best observed objective value after n evaluations is defined as

$$J_{\min}^{(n)} = \min_{1 \le i \le n} J_i. \tag{3.17}$$

For a candidate parameter vector $\mathbf{x}$, the improvement over the current best observed value is

$$I(\mathbf{x}) = \max\left(J_{\min}^{(n)} - J(\mathbf{x}), 0\right). \tag{3.18}$$

Since the true objective value $f(\mathbf{x})$ is unknown before the expensive Simscape evaluation, the expected improvement is calculated using the posterior predictive

distribution of the Gaussian-process surrogate:

$$EI(\mathbf{x}) = E\left[I(\mathbf{x})\right] = E\left[\max\left(J_{\min}^{(n)} - J(\mathbf{x}), 0\right)\right]. \tag{3.19}$$

The next parameter vector proposed by the Bayesian optimization engine is therefore selected as

$$\mathbf{x}_{n+1} = \arg\max_{\mathbf{x}\in\Omega} EI(\mathbf{x}). \tag{3.20}$$

This acquisition step is responsible for balancing exploitation and exploration. A candidate parameter set may receive a high acquisition value either because its predicted fitting cost is low, or because its predictive uncertainty is large. Therefore, the acquisition function guides the optimizer to evaluate parameter sets that are most likely to improve the current best result.

The proposed candidate parameter vector is then sent to the Simscape PEMFC system model. In the simulation step, the model workspace is updated using the candidate parameter values, and the PEMFC system is simulated under the predefined power profile. The resulting simulated efficiency curve is extracted and compared with the reference efficiency–power curve.

After the objective function evaluation based on Equation 3.13, a new observed parameter–cost pair is obtained:

$$(\mathbf{x}_{n+1}, J_{n+1}), \tag{3.21}$$

where

$$J_{n+1} = J(\mathbf{x}_{n+1}). \tag{3.22}$$

This new observation is added to the evaluated dataset:

$$D_{n+1} = D_n \cup \{(\mathbf{x}_{n+1}, J_{n+1})\}. \tag{3.23}$$

The Gaussian-process surrogate model is then updated using the enlarged dataset D_{n+1}. This updated surrogate model is used in the next iteration to compute a new acquisition function and propose the next candidate parameter vector. This loop is repeated for 200 iterations. The best observed parameter vector is finally selected as

$$\mathbf{x}^* = \arg\min_{\mathbf{x}_i\in D_N} J_i, \tag{3.24}$$

where N is the total number of completed objective evaluations. The identified parameter vector $\mathbf{x}^*$ is then used as the calibrated stack-level parameter set for the PEMFC Simscape model.

3.3 Machine learning model of operating condition indicators

The hydrogen excess ratio (HER), oxygen excess ratio (OER), and membrane water content λ_w are selected in this work as the main internal operating condition indicators of the PEM fuel cell stack. These variables describe the reactant supply condition and membrane hydration state, and are therefore closely related to degradation-relevant phenomena such as hydrogen starvation, oxygen starvation, membrane dehydration, and flooding. However, in practical fuel cell systems, these internal variables are difficult to obtain directly from standard real-time sensors. To address this limitation, this work constructs a supervised machine-learning model for internal-state estimation. The calibrated Simscape PEM fuel cell system model is first used as a high-fidelity data-generation model. Given a fuel-cell power-demand profile, the physical model generates the corresponding HER, OER, and λ_w responses. A machine-learning model is then trained to approximate the mapping from externally available power-demand features to these internal operating indicators. It should be noted that the machine-learning model is not used as a black-box lifetime predictor. Instead, it serves as a computationally efficient surrogate estimator for physically interpretable internal states.

The overall mapping learned by the machine-learning model can be expressed as

$$\hat{\mathbf{y}}_k = f_{\mathrm{ML}}\left(\mathbf{x}_k^{(h)}\right), \tag{3.25}$$

where $\mathbf{x}_k^{(h)}$ is the input feature vector constructed from the fuel-cell power demand and its historical information within a time window of length h, and f_{ML} denotes the trained regression model. The output vector is defined as

$$\mathbf{y}_k = [HER_k,\ OER_k,\ \lambda_{w,k}]^{\mathrm{T}}. \tag{3.26}$$

3.3.1 Dataset generation

The power dataset used for machine-learning model training is generated from twenty operating segments, each with a duration of 300 s. These segments are extracted from two groups of fuel-cell power-demand trajectories. Each segment represents the power command assigned by the energy management system (EMS) to the fuel cell system. For each power-demand segment, the calibrated physics-based PEMFC model is simulated, and the corresponding HER, OER, and membrane water content responses are recorded as supervised-learning targets.

The two groups of operating segments are used for different validation purposes. Group A, shown in Figure 3.6, is used as the primary dataset for model training, internal testing, and segment-level cross-validation. Specifically, the first seven segments in Group A are used for training, while the remaining three segments are used as an internal block-based test set. This test evaluates whether the model can predict unseen temporal blocks from the same group of operating trajectories. Group B, shown in Figure 3.7, is reserved as an independent external validation set and is not used during model training. This external

validation evaluates whether the learned mapping can be transferred to a different group of operating trajectories without retraining.

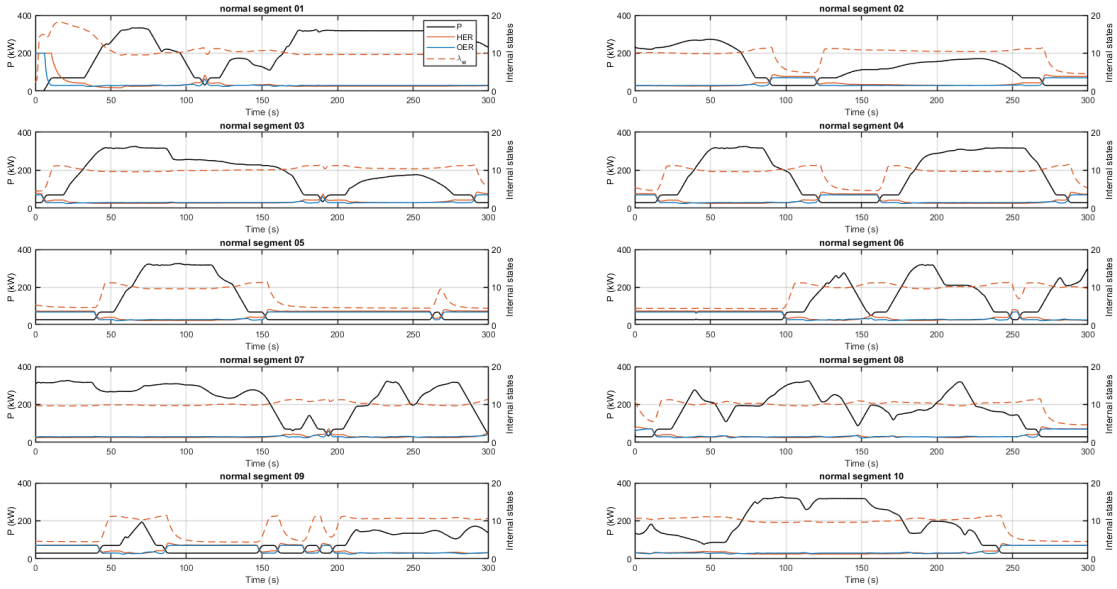


Figure 3.6: Group A operating profiles and corresponding internal state responses.

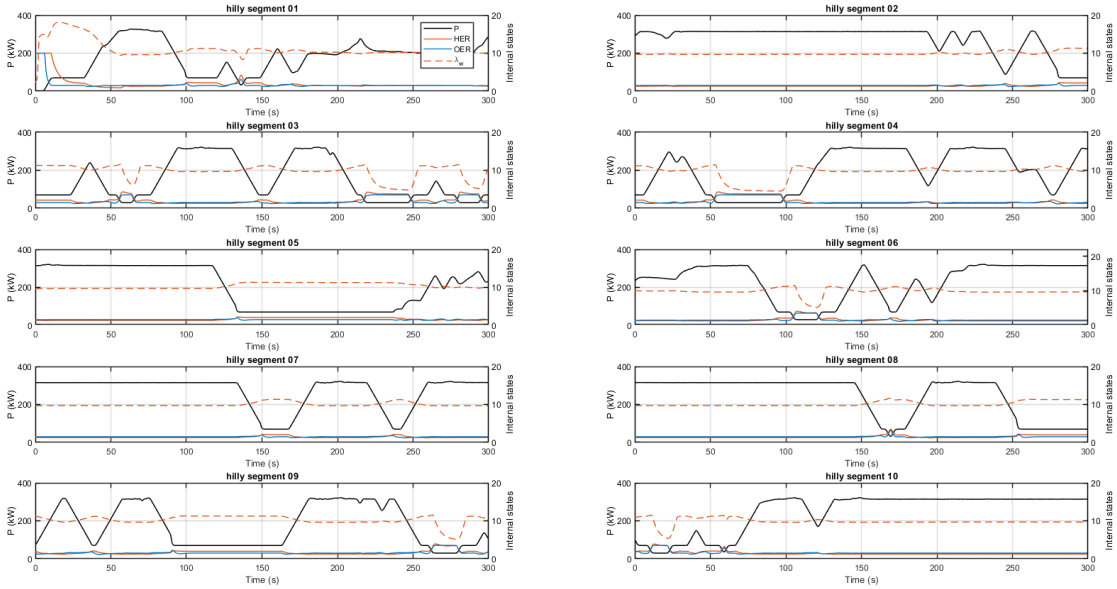


Figure 3.7: Group B operating profiles and corresponding internal state responses.

Furthermore, a segment-level K-fold cross-validation procedure is performed using only the Group A segments. The cross-validation is conducted at the segment level rather

than at the individual sample level because samples extracted from the same operating segment are temporally correlated and share the same load-history pattern. A random sample-level split would therefore allow neighbouring samples from the same segment to appear in both the training and validation sets, leading to information leakage and an over-optimistic estimate of model generalization. For each fold, a subset of Group A segments is held out for validation, while the remaining segments are used for training. The independent Group B segments are excluded from the K-fold procedure and are used only for external validation. The dataset partition can be summarized as

$$D_A = D_A^{\text{train}} \cup D_A^{\text{test}}, \qquad D_A^{\text{train}} \cap D_A^{\text{test}} = \emptyset, \tag{3.27}$$

and

$$D_A \cap D_B = \emptyset, \tag{3.28}$$

where D_A denotes the Group A dataset and D_B denotes the independent external validation dataset.

3.3.2 Input feature construction

The input features are extracted from the externally available fuel-cell power-demand signal. At the current time step k, the fuel-cell power demand is denoted as

$$P_k = P_{\text{fc}}(t_k). \tag{3.29}$$

The instantaneous power variation rate is approximated by

$$\dot{P}_k = \frac{P_k - P_{k-1}}{\Delta t}, \tag{3.30}$$

where Δt is the sampling interval.

To account for the effect of recent operating history, a moving historical window is defined as

$$W_k^{(h)} = \{ j \mid t_k - h \le t_j \le t_k \}, \tag{3.31}$$

where h is the window length. Within this historical window, statistical and transient features are extracted to describe the recent power level, power range, fluctuation intensity, transient severity, operating-region occupancy, and slope direction.

The complete input feature vector for a historical window of length h is written as

$$\begin{aligned} \mathbf{x}_k^{(h)} = \big[& P_k,\ \dot{P}_k,\ \overline{P}_k^{(h)},\ P_{\text{min},k}^{(h)},\ P_{\text{max},k}^{(h)},\ \sigma_{P,k}^{(h)}, \\ & \overline{\dot{P}}_k^{(h)},\ \sigma_{\dot{P},k}^{(h)},\ \overline{|\dot{P}|}_k^{(h)},\ |\dot{P}|_{\text{max},k}^{(h)}, \\ & r_{\text{low},k}^{(h)},\ r_{\text{high},k}^{(h)},\ r_{\text{pos},k}^{(h)},\ r_{\text{neg},k}^{(h)} \big]^{\mathrm{T}}. \end{aligned} \tag{3.32}$$

The meanings of these candidate input features are summarized in Table 3.2. The low-power ratio $r_{\text{low},k}^{(h)}$ and high-power ratio $r_{\text{high},k}^{(h)}$ describe how long the system operates in low- and high-power regions within the historical window. The positive- and negative-slope

ratios describe the frequency of load-increase and load-decrease events. These features are included because HER , OER are sensitive to the mismatch between reactant consumption and gas-supply response during load transients, while λ_w is affected by slower hydration and water-removal dynamics.

Table 3.2: Input features used for internal-state prediction.

Feature name	**Symbol / expression**	**Description**
Current power	P_k	Instantaneous fuel-cell power demand at the current time step.
Current power slope	$\dot{P}_k$	Instantaneous rate of change of the power demand.
Mean power	$\overline{P}_k^{(h)}$	Mean value of the power demand within the historical window.
Minimum power	$P_{\min,k}^{(h)}$	Minimum power demand within the historical window.
Maximum power	$P_{\max,k}^{(h)}$	Maximum power demand within the historical window.
Standard deviation of power	$\sigma_{P,k}^{(h)}$	Standard deviation of the power demand within the historical window.
Mean power slope	$\overline{\dot{P}}_k^{(h)}$	Mean value of the power variation rate within the historical window.
Standard deviation of power slope	$\sigma_{\dot{P},k}^{(h)}$	Standard deviation of the power variation rate within the historical window.
Mean absolute power slope	$\overline{\lvert\dot{P}\rvert}_k^{(h)}$	Mean absolute value of the power variation rate within the historical window.
Maximum absolute power slope	$\lvert\dot{P}\rvert_{\max,k}^{(h)}$	Maximum absolute value of the power variation rate within the historical window.
Low-power ratio	$r_{\text{low},k}^{(h)}$	Fraction of samples in the historical window satisfying $P < 60$ kW.
High-power ratio	$r_{\text{high},k}^{(h)}$	Fraction of samples in the historical window satisfying $P > 200$ kW.
Positive-slope ratio	$r_{\text{pos},k}^{(h)}$	Fraction of samples in the historical window satisfying $\dot{P} > 0.5$ kW/s.
Negative-slope ratio	$r_{\text{neg},k}^{(h)}$	Fraction of samples in the historical window satisfying $\dot{P} < -0.5$ kW/s.

Based on these candidate features, six feature configurations are evaluated. The first configuration uses only the instantaneous power P_k. The second configuration uses both P_k and $\dot{P}_k$. The remaining four configurations use the complete history-dependent feature vector in Equation (3.32), with historical window lengths of 5 s, 10 s, 20 s, and 40 s,

respectively. This comparison is designed to investigate whether the internal operating condition indicators can be estimated from the instantaneous operating point alone, or whether recent load-history information is necessary to capture the dynamic behavior of reactant supply and membrane hydration.

3.3.3 Supervised regression formulation

For a training dataset containing N samples, the supervised learning dataset is defined as

$$D_{\mathrm{ML}} = \left\{ \left(\mathbf{x}_k^{(h)}, \mathbf{y}_k \right) \right\}_{k=1}^{N}. \tag{3.33}$$

The model parameters are obtained by minimizing the empirical prediction error:

$$\theta^* = \arg\min_{\theta} \frac{1}{N} \sum_{k=1}^{N} \left\| \mathbf{y}_k - f_{\mathrm{ML}} \left(\mathbf{x}_k^{(h)}; \theta \right) \right\|_2^2. \tag{3.34}$$

In implementation, the three internal variables HER, OER, and λ_w can be trained as separate regression targets. This makes it possible to evaluate the prediction accuracy and feature sensitivity of each internal state independently. The prediction errors are evaluated using standard regression metrics, including the root mean square error (RMSE), mean absolute error (MAE), and coefficient of determination R^2.

3.3.4 Regression models

The purpose of the regression-model comparison is to identify a suitable data-driven surrogate for estimating the internal operating condition indicators. In this task, the input variables are feature-based descriptions of the fuel-cell power demand, and the relationship between the load features and the internal states is expected to be nonlinear. For HER and OER, a rapid increase in power demand immediately increases the reactant consumption rate, while the gas-supply dynamics may respond with a delay. Therefore, the same power level can lead to different HER and OER values depending on whether the system has reached a quasi-steady condition or is experiencing a transient load increase. For membrane water content, the dependence on recent history is even more pronounced, because water generation, membrane hydration, water removal, and transport processes have slower dynamics than the electrical load. Therefore, three nonlinear regression strategies are evaluated in this work: Gaussian process regression (GPR), support vector regression (SVR), and random forest regression (RF).

Gaussian process regression

GPR is included because it provides a probabilistic surrogate for internal-state estimation. It is used to test whether a smooth uncertainty-aware regression model is suitable for mapping power-demand features to internal fuel-cell states.

For a scalar internal-state target, the unknown regression function is modelled as

$$f(\mathbf{x}) \sim GP\left(m(\mathbf{x}),\, k(\mathbf{x}, \mathbf{x}')\right), \tag{3.35}$$

where $m(\mathbf{x})$ is the mean function and $k(\mathbf{x}, \mathbf{x}')$ is the covariance function. In the present application, $\mathbf{x}$ represents the load-history feature vector, and $f(\mathbf{x})$ represents the corresponding internal-state response.

After conditioning on the training data, the prediction for a new input $\mathbf{x}_*$ is expressed as

$$y_* \mid \mathbf{x}_*, D_{\mathrm{ML}} \sim \mathcal{N}\left(\mu_*(\mathbf{x}_*),\, \sigma_*^2(\mathbf{x}_*)\right). \tag{3.36}$$

The predicted mean $\mu_*(\mathbf{x}_*)$ is used as the estimated HER, OER, or λ_w, while the predictive variance $\sigma_*^2(\mathbf{x}_*)$ indicates the confidence of the estimator. Therefore, GPR is particularly useful for evaluating whether the learned mapping remains reliable when applied to the independent Group B operating profiles.

Support vector regression

SVR is also included as a kernel-based nonlinear regression baseline for the internal-state estimation task. Since the relationship between these load-history features and HER, OER, or λ_w is not expected to be purely linear, SVR is used to examine whether this nonlinear mapping can be represented through kernel-based regression. It provides a useful comparison model for evaluating whether the internal fuel-cell states can be estimated from power-demand features without relying on a probabilistic surrogate formulation.

For a scalar target variable, the SVR prediction can be written as

$$\hat{y}_k = f_{\mathrm{SVR}}(\mathbf{x}_k) = \mathbf{w}^{\mathrm{T}}\phi(\mathbf{x}_k) + b, \tag{3.37}$$

where $\mathbf{x}_k$ is the load-feature vector at time step k, $\hat{y}_k$ is the predicted internal-state value, $\phi(\mathbf{x}_k)$ maps the original input features into a higher-dimensional feature space, $\mathbf{w}$ is the weight vector, and b is the bias term.

The nonlinear capability of SVR is introduced through the kernel function:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^{\mathrm{T}}\phi(\mathbf{x}_j). \tag{3.38}$$

The kernel function measures the similarity between two load-history feature vectors after they are implicitly mapped into a higher-dimensional feature space. In the present application, this means that two operating points with similar power level, power ramp rate, and recent load-history characteristics are expected to have similar HER, OER, or λ_w responses.

In this way, SVR is used to test whether the nonlinear dependence of internal fuel-cell states on operating-point and load-history features can be captured by a kernel-based regression model. Its performance is then compared with GPR and RF under the same feature configurations to evaluate the suitability of different regression strategies for the data-driven internal-state estimator.

Random forest regression

Unlike GPR and SVR, which rely on kernel-based smoothness assumptions, RF learns the input-output relationship through an ensemble of decision trees. This makes it suitable for capturing threshold-like effects and interactions between different operating features.

For a given input feature vector $\mathbf{x}_k$, the RF prediction is obtained by averaging the outputs of M regression trees:

$$\hat{y}_k = f_{\mathrm{RF}}(\mathbf{x}_k) = \frac{1}{M} \sum_{m=1}^{M} T_m(\mathbf{x}_k), \tag{3.39}$$

where $T_m(\mathbf{x}_k)$ is the prediction of the m-th tree. This structure is useful for the present problem because different internal states may depend on different combinations of load features. For example, HER and OER may be more sensitive to instantaneous power and positive power ramps, while λ_w may be more strongly affected by historical power level, fluctuation intensity, and the duration of high-power operation.

Another reason for including RF is its ability to provide predictor-importance information. This allows the contribution of different load features to be analysed after training. Therefore, RF is not only used as a prediction model, but also as a tool for interpreting which external power-demand features are most relevant to estimating the internal operating condition indicators.

By comparing these models under the six feature configurations, the thesis evaluates two questions. First, it examines whether instantaneous power alone is sufficient for internal-state estimation, or whether historical load features improve the prediction of HER, OER, and λ_w. Second, it evaluates which regression strategy is most suitable for constructing the data-driven component of the grey-box degradation modelling framework.

3.3.5 Model robustness evaluation

In addition to cross-validation accuracy on Group A and external generalization on Group B, two supplementary evaluations are conducted to assess the robustness of the trained internal-state estimators under conditions that deviate from the nominal training setting.

Monte Carlo noise sensitivity analysis

In practical fuel cell systems, the power-demand signal used as input to the estimators may be subject to measurement uncertainty. To quantify the sensitivity of the trained models to such uncertainty, a Monte Carlo simulation is performed on the Group B operating profiles. Zero-mean Gaussian noise is independently added to the power signal at each time step before feature extraction, with standard deviations set to 1%, 2%, and 5% of the maximum power P_{max}:

$$\tilde{P}_k = P_k + \epsilon_k, \quad \epsilon_k \sim \mathcal{N}(0,\, \sigma^2), \quad \sigma \in \{0.01,\, 0.02,\, 0.05\} \cdot P_{\mathrm{max}}. \tag{3.40}$$

For each noise level, 50 independent trials are conducted. At each trial, the noisy power signal is used to extract input features, which are then passed to the trained estimator. The resulting RMSE and R^2 values are recorded across trials, and their mean and standard deviation are reported to characterise prediction degradation as a function of input noise level.

Robustness under unmodelled operating conditions

The machine-learning estimators are trained using simulation data generated under a fixed set of operating conditions, with cathode relative humidity and stack temperature held at their nominal calibrated values. In practice, these quantities may vary during vehicle operation. Since neither cathode relative humidity nor stack temperature is included in the input feature set, the estimators cannot directly compensate for their influence. To assess the sensitivity of the estimators to such variation, two parameter sweeps are conducted using the calibrated Simscape model.

The first sweep varies cathode relative humidity across three levels,

$$\mathrm{RH}_{\mathrm{cat}} \in \{0.65,\ 0.75,\ 0.85\}, \tag{3.41}$$

at a fixed stack temperature of $T = 25\,^{\circ}\mathrm{C}$. The second sweep varies stack temperature across three levels,

$$T \in \{15,\ 25,\ 35\}\,^{\circ}\mathrm{C}, \tag{3.42}$$

at a fixed cathode relative humidity of $\mathrm{RH}_{\mathrm{cat}} = 0.78$.

For each condition, the Group B power profiles are applied to the modified simulation model to generate updated HER, OER, and membrane water content responses, which serve as the reference targets. The trained estimators, which receive only fuel-cell power demand features as inputs, are then evaluated against these updated references. The resulting signed relative error distributions and R^2 values are compared across conditions to identify which internal states are most sensitive to unmodelled operating variables, and to establish the operational validity range of the proposed estimators.

3.4 Grey-box degradation index formulation

The purpose of this final stage is to connect the estimated internal operating condition indicators with a measurable degradation-related quantity. Although HER, OER, and membrane water content provide physically meaningful information about reactant supply and water-management conditions, they do not directly represent the voltage degradation rate of the stack. Therefore, a degradation index is constructed to aggregate these internal-state trajectories and power-history features into a scalar degradation severity score.

3.4.1 Reference degradation dataset

To calibrate the proposed grey-box degradation index, a reference degradation dataset is constructed using dynamic operating profiles with experimentally measured degradation

labels. For the i-th operating profile, the experimentally obtained ground-truth voltage loss rate is denoted as

$$y_i = \dot{V}_{\text{loss},i}, \tag{3.43}$$

where y_i is expressed in $\mu\text{V}/\text{h}$. These experimentally measured voltage loss rates are used as the ground-truth calibration targets for the grey-box degradation index. They are used to calibrate and evaluate its predicted degradation severity.

The dataset is divided into three groups, as summarized in Table 3.3. The first group contains symmetric two-level cycling profiles with different low-power and high-power levels under two ramp-rate settings. The second group consists of average-up-down dynamic cycling profiles, where the average operating level and the upward/downward power excursions are varied. The third group evaluates symmetric cycling profiles with different hold times. Together, these profiles cover different load-amplitude, ramp-rate, power-excursion, and hold-time conditions.

Table 3.3: Summary of the reference degradation dataset used for calibration.

Group	Profile type	Varied quantities	Number of cases
SYM	Symmetric two-level cycling	Low/high power, ramp rate	18
AVG	Average-up-down cycling	Start, step-up, step-down power	17
SYM_VAR	Hold-time variation	Low/high power, hold time	18
Total	–	–	53

For each operating profile, the ground-truth voltage loss rate is obtained from experimental degradation data and used as the calibration target. The corresponding power-demand profile is used as the input to the trained internal-state estimator introduced in the previous section. This estimator predicts the time-series trajectories of HER, OER and membrane water content. Together with the power-demand profile, these predicted internal-state trajectories are used to calculate the degradation-related severity terms introduced in the following subsection. The complete list of operating profiles and their ground-truth voltage loss rates is provided in Appendix **??**.

3.4.2 Definition of degradation-related severity terms

To construct a physically interpretable degradation index, seven degradation-related severity terms are defined in this work. These terms are designed to describe different types of degradation-prone operation in the PEM fuel cell system. The first four terms are derived from the internal operating state variables, which describe reactant starvation risk and membrane hydration imbalance. The remaining three terms consists of external-load severity terms extracted directly from the fuel-cell power-demand profile. By this form, the degradation index combines excitation-level descriptors and response-level indicators. The final contribution of each candidate term is determined by a set of calibrated weights. This allows the calibration process to identify which internal-response and external-load terms are most relevant for explaining the reference degradation data.

The first severity term describes hydrogen starvation risk. Since hydrogen starvation is associated with insufficient anode reactant supply, the penalty is activated only when

the hydrogen excess ratio falls below a reference value $\lambda_{H_2,\text{ref}}$. The hydrogen-starvation severity term is therefore defined as

$$J_H = \frac{1}{T}\int_0^T \max\left(0, \lambda_{H_2,\text{ref}} - HER(t)\right)^2 \, dt. \tag{3.44}$$

This formulation penalizes both the magnitude and duration of insufficient hydrogen supply. If $HER(t)$ remains above the reference level, the corresponding contribution is zero. If $HER(t)$ decreases below the reference level, a larger and longer deviation gives a larger degradation-related penalty.

Similarly, the oxygen-starvation severity term is defined based on the oxygen excess ratio:

$$J_O = \frac{1}{T}\int_0^T \max\left(0, \lambda_{O_2,\text{ref}} - OER(t)\right)^2 \, dt, \tag{3.45}$$

where $\lambda_{O_2,\text{ref}}$ is the reference oxygen excess ratio. This term represents the risk associated with insufficient oxygen supply on the cathode side. During rapid load increases, the stack current and oxygen consumption may increase faster than the air supply system can respond, which can lead to a temporary decrease in $OER(t)$. Therefore, this term captures the degradation-prone operating condition associated with transient oxygen starvation.

The third severity term describes the deviation of membrane water content from a reference hydration level. The membrane water-content state severity is defined as

$$J_{m,\text{state}} = \frac{1}{T}\int_0^T \left(\lambda_w(t) - \lambda_{w,\text{ref}}\right)^2 \, dt, \tag{3.46}$$

where $\lambda_{w,\text{ref}}$ is the reference membrane water content. This term penalizes operation away from the preferred membrane hydration condition. A large positive or negative deviation may indicate water-management imbalance, such as membrane dehydration or excessive water accumulation. Therefore, this term is included to represent degradation risk related to membrane hydration state.

In addition to the absolute hydration level, the dynamic variation of membrane water content is also considered. Frequent hydration and dehydration cycling can increase mechanical and chemical stress in the membrane. To represent this effect, the membrane hydration-cycling severity is defined as

$$J_{m,\text{cycle}} = \frac{1}{T}\int_0^T \left(\frac{d\lambda_w(t)}{dt}\right)^2 \, dt. \tag{3.47}$$

This term becomes large when the membrane water content changes rapidly during dynamic operation. It therefore represents the severity of membrane hydration cycling, rather than the absolute hydration level itself.

The fifth term represents high-power operation severity. High fuel-cell power demand is generally associated with high current density, stronger reactant consumption, larger voltage losses, and more demanding water and thermal management. In this work, the

high-power severity term is defined as

$$J_{\text{highP}} = \frac{1}{T}\int_0^T \max\left(0, P(t) - P_{\text{high,ref}}\right)^2 \, dt, \tag{3.48}$$

where $P_{\text{high,ref}}$ is the reference high-power threshold. This term is only activated when the demanded power exceeds the reference threshold. It therefore measures the duration and magnitude of operation in the high-power region.

The sixth term describes power-cycling severity. Dynamic load changes can cause transient mismatch between reactant consumption and supply, and can also induce variations in water generation, membrane hydration, and thermal conditions. The power-cycling severity is defined as

$$J_{P,\text{cycle}} = \frac{1}{T}\int_0^T \left(\frac{dP(t)}{dt}\right)^2 \, dt. \tag{3.49}$$

This term penalizes rapid power variations. Compared with the high-power term, it does not depend on the absolute power level only, but instead represents the transient severity of the load profile.

Finally, the power-amplitude severity term is introduced to describe the overall power variation range within an operating profile:

$$J_{\text{amp}} = \left(P_{\text{max}} - P_{\text{min}}\right)^2, \tag{3.50}$$

where P_{max} and P_{min} are the maximum and minimum fuel-cell power demands within the profile. This term represents the severity of large-amplitude load cycling. It complements $J_{P,\text{cycle}}$, because two profiles may have similar instantaneous power ramp rates but different overall power-amplitude ranges.

The complete set of degradation-related severity terms can therefore be written as

$$\mathbf{J} = \left[J_H,\ J_O,\ J_{m,\text{state}},\ J_{m,\text{cycle}},\ J_{\text{highP}},\ J_{P,\text{cycle}},\ J_{\text{amp}}\right]^T. \tag{3.51}$$

These seven terms provide a structured description of degradation-prone operation from both internal-state and load-history perspectives. The internal-state-based terms preserve the physical connection to reactant starvation and membrane water management, while the power-profile-based terms account for external operating severity. The following subsection explains how these terms are normalized and combined into a scalar grey-box degradation score.

Since the seven degradation-related severity terms defined have different physical meanings, numerical ranges, and units, these terms cannot be directly added together before normalization. To make the different severity terms comparable, each term is first normalized over the complete calibration dataset.

For the j-th degradation-related severity term, the raw value of the i-th operating profile

is denoted by $J_{i,j}$. The normalized value is calculated using min–max normalization:

$$J_{i,j} = \frac{J_{i,j} - J_{j,\text{min}}}{J_{j,\text{max}} - J_{j,\text{min}}}, \tag{3.52}$$

where $J_{j,\text{min}}$ and $J_{j,\text{max}}$ are the minimum and maximum values of the j-th severity term among all calibration profiles. After normalization, each severity term is scaled to a comparable range between zero and one. A larger normalized value indicates a more severe operating condition with respect to that specific degradation-related term.

The normalized degradation-term vector of the i-th operating profile is written as

$$\tilde{\mathbf{J}}_i = \left[J_{H,i},\ J_{O,i},\ J_{m,\text{state},i},\ J_{m,\text{cycle},i},\ J_{\text{highP},i},\ J_{P,\text{cycle},i},\ J_{\text{amp},i}\right]^T. \tag{3.53}$$

The final degradation score is then constructed as a weighted sum of these normalized severity terms:

$$J_{\text{deg},i} = \mathbf{w}^T \tilde{\mathbf{J}}_i, \tag{3.54}$$

where the weight vector is defined as

$$\mathbf{w} = \left[w_H,\ w_O,\ w_{m,\text{state}},\ w_{m,\text{cycle}},\ w_{\text{highP}},\ w_{P,\text{cycle}},\ w_{\text{amp}}\right]^T. \tag{3.55}$$

The weights determine the relative contribution of each candidate severity term to the final degradation index. Since the seven terms are introduced as candidate descriptors rather than being assumed to have equal importance, their weights are identified during the calibration process. To preserve the interpretability of the weighted sum, the weights are constrained to be non-negative and to sum to one:

$$w_j \geq 0, \qquad \sum_{j=1}^{7} w_j = 1. \tag{3.56}$$

In implementation, this constraint is enforced using a softmax parameterization. Let $\mathbf{z} = [z_1, z_2, \ldots, z_7]^T$ be an unconstrained vector of weight parameters. The corresponding normalized weights are calculated as

$$w_j = \frac{\exp(z_j)}{\sum_{l=1}^{7} \exp(z_l)}, \qquad j = 1, \ldots, 7. \tag{3.57}$$

This formulation guarantees that all weights are positive and that their sum is equal to one. As a result, the calibrated weights can be interpreted as the relative importance of the different degradation-related severity terms within the proposed grey-box degradation index.

It should be noted that J_{deg} is a dimensionless degradation severity score, rather than a direct physical measurement of voltage loss. The score is designed to preserve the relative degradation tendency among different operating profiles. A larger value of J_{deg} indicates that the corresponding operating profile is expected to be more degradation-

prone according to the combined internal-response and external-load severity information. The quantitative mapping from J_{deg} to the reference voltage loss rate is introduced in the calibration step described in the following subsection.

3.4.3 Calibration of weights and thresholds

After the degradation-related severity terms and the weighted degradation score have been defined, the remaining task is to calibrate the reference thresholds and the term weights using the reference voltage-loss-rate data. The purpose of this calibration step is to make the grey-box degradation index quantitatively consistent with the reference degradation labels.

The optimized calibration parameter vector is defined as

$$\boldsymbol{\theta} = \left[\lambda_{H_2,\text{ref}},\ \lambda_{O_2,\text{ref}},\ \lambda_{w,\text{ref}},\ P_{\text{high,ref}},\ z_1, \ldots, z_7\right]^T, \tag{3.58}$$

where the first four variables are the reference thresholds used in the degradation-related severity terms, and $z_1, \ldots, z_7$ are the softmax logits used to generate the degradation-term weights. The final weights are obtained using the softmax parameterization described in Equation 3.57, which guarantees that all weights are non-negative and sum to one.

For a given calibration parameter vector $\boldsymbol{\theta}$, the seven severity terms are recalculated, normalized, and combined into the dimensionless degradation score $J_{\text{deg},i}$. Since $J_{\text{deg},i}$ is not itself a voltage-loss quantity, a calibration mapping is introduced between the normalized degradation score and the normalized reference voltage loss rate:

$$\hat{\tilde{y}}_i = \beta_0 + \beta_1 \tilde{J}_{\text{deg},i}, \tag{3.59}$$

where $\tilde{J}_{\text{deg},i}$ and $\tilde{y}_i$ are the min–max-normalized degradation score and reference voltage loss rate, respectively. For each candidate $\boldsymbol{\theta}$, the coefficients β_0 and β_1 are obtained by ordinary least squares. Therefore, the calibration does not force the dimensionless score itself to have the same unit as the voltage loss rate; instead, it identifies whether the score can be mapped consistently to the reference degradation value.

The main calibration loss is defined as the mean squared error between the normalized prediction and the normalized reference voltage loss rate:

$$L_{\text{fit}} = \frac{1}{N} \sum_{i=1}^{N} \left(\hat{\tilde{y}}_i - \tilde{y}_i\right)^2, \tag{3.60}$$

where N is the number of calibration profiles. To maintain numerical stability and avoid degenerate solutions, two small regularization terms are also included. The final calibration objective is written as

$$L(\boldsymbol{\theta}) = L_{\text{fit}} + R_w + 0.01 R_{\text{spread}}. \tag{3.61}$$

In which R_w is a small weight regularization term, and R_{spread} prevents the calibrated degradation scores from collapsing into a nearly constant value.

The calibration problem is formulated as a bounded nonlinear optimization problem:

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} L(\boldsymbol{\theta}) \tag{3.62}$$

subject to

$$\boldsymbol{\theta}_{lb} \leq \boldsymbol{\theta} \leq \boldsymbol{\theta}_{ub}. \tag{3.63}$$

The bounds of the calibration variables are summarized in Table 3.4.

Table 3.4: Bounds of the calibration variables in the degradation-index optimization.

Variable	Lower bound	Upper bound
$\lambda_{H_2,\text{ref}}$	1.10	1.90
$\lambda_{O_2,\text{ref}}$	1.20	3.50
$\lambda_{w,\text{ref}}$	8.00	20.00
$P_{\text{high,ref}}$	80.0 kW	280.0 kW
$z_j,\ j = 1, \ldots, 7$	-6	6

The bounds for $\lambda_{H_2,\text{ref}}$ and $\lambda_{O_2,\text{ref}}$ are selected to cover both near-stoichiometric and over-supply operation. In PEMFC systems, excess ratios should remain above one to avoid reactant starvation, and previous studies have reported typical over-stoichiometric operation above approximately 1.2 on the anode side and 1.5 on the cathode side [34]. The membrane-water-content bound $[8, 20]$ covers the main simulated hydration range, while the high-power threshold bound $[80, 280]$ kW covers the medium- and high-load regions of the investigated profiles. The logits z_j are bounded within $[-6, 6]$ to keep the numerical search stable, while the final weights are still obtained through softmax normalization.

The bounded nonlinear optimization problem is solved using the SQP algorithm implemented in MATLAB. In the standard fmincon form, the linear inequality constraints, linear equality constraints, and nonlinear constraints are left empty. Therefore, the only explicit constraints are the lower and upper bounds in Table 3.4. At each iteration, SQP constructs a local quadratic approximation of the nonlinear objective function and solves the corresponding subproblem to obtain an update direction. Since only bound constraints are imposed in this calibration problem, the update direction satisfies

$$\boldsymbol{\theta}_{lb} - \boldsymbol{\theta}^{(r)} \leq \mathbf{d}^{(r)} \leq \boldsymbol{\theta}_{ub} - \boldsymbol{\theta}^{(r)}. \tag{3.64}$$

The parameter vector is updated according to

$$\boldsymbol{\theta}^{(r+1)} = \boldsymbol{\theta}^{(r)} + \alpha^{(r)}\mathbf{d}^{(r)}, \tag{3.65}$$

where $\alpha^{(r)}$ is the step length selected by the optimizer.

To reduce the sensitivity to local minima, a multi-start strategy is used. The first run is initialized from a manually selected parameter vector, and the remaining runs start from randomly sampled points within the bounded search space. The solution with the lowest final objective value is selected as the final calibrated parameter set.

Chapter 4

Results and Analysis

This chapter presents the main results and analysis of the study. First, the parameter identification process of the FCS physical model is presented, and the identified results are used to demonstrate the realism and validity of the Simscape-based fuel cell system model adopted in this thesis. Then, different machine learning methods and model parameters are compared to evaluate their influence on the prediction accuracy of the internal operating state variables. Finally, the proposed degradation model is validated using real experimental data, in order to assess its ability to represent degradation trends under practical operating conditions.

4.1 Parameter calibration results

Figure 4.1 shows the convergence history of the Bayesian optimization process. The blue curve represents the objective value obtained from each valid Simscape evaluation, while the red curve represents the best-so-far objective value. Failed simulations are assigned a large penalty value and are marked separately in the figure.

The best-so-far objective value decreases rapidly during the early stage of the optimization, indicating that Bayesian optimization is able to efficiently identify a promising low-cost region in the high-dimensional parameter space. After approximately 60 function evaluations, the best-so-far objective reaches a plateau, and no significant improvement is observed in the remaining evaluations. The final best objective value is

$$J_{\text{best}} = 8.31 \times 10^{-5}. \tag{4.1}$$

which corresponds to a root-mean-square efficiency error of approximately 0.0091, or about 0.91 percentage points.

The calibrated model is further evaluated using the error between the normalized simulated steady-state efficiency and the normalized reference efficiency. Instead of directly presenting the original efficiency–power curves, both efficiency series are normalized before comparison to avoid disclosing proprietary efficiency data. As shown in Figure 4.2, the normalized efficiency error remains small over most of the investigated power range, indicating that the identified parameter set captures the main efficiency trend

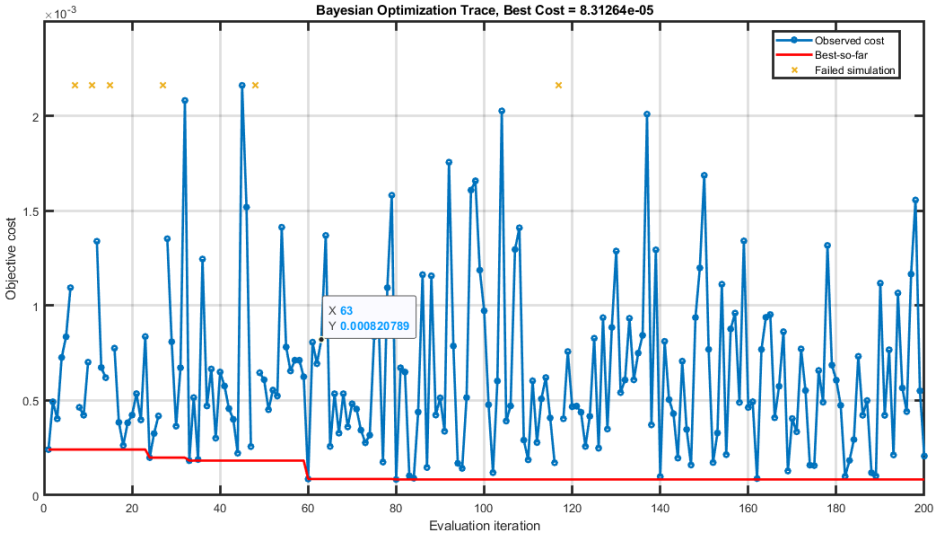


Figure 4.1: Bayesian optimization convergence history for Simscape parameter identification

of the fuel cell system. The agreement is particularly good in the medium- and high-power regions, where the error stays close to zero. A larger deviation is observed in the low-power region, suggesting a residual mismatch between the simplified Simscape model and the reference system behaviour. This discrepancy may be attributed to the limited flexibility of the simplified balance-of-plant representation, auxiliary power correction, or the weighting strategy used in the objective function.

To qualitatively assess the exploration behavior of the Bayesian optimization process, the evaluated parameter sets are projected into two-dimensional spaces using Principal Component Analysis (PCA) and t-distributed Stochastic Neighbor Embedding (t-SNE), as shown in Figure 4.3. Before projection, all optimized parameters are normalized according to their predefined lower and upper bounds:

$$\tilde{p}_j = \frac{p_j - p_{j,\min}}{p_{j,\max} - p_{j,\min}}, \tag{4.2}$$

where p_j is the value of the j-th parameter, and $p_{j,\min}$ and $p_{j,\max}$ are its lower and upper bounds, respectively.

The PCA and t-SNE projections show that the evaluated samples are distributed over multiple regions of the normalized parameter space, indicating that the optimization process maintained sufficient sampling diversity during the search. Meanwhile, the best parameter set appears in a low-cost region identified after a broad exploration of candidate combinations. This behavior is consistent with the expected mechanism of Bayesian optimization: early evaluations provide broader exploration of the feasible domain, while later evaluations increasingly exploit promising regions suggested by the surrogate model.

The final identified parameters are summarized in Table 4.1.

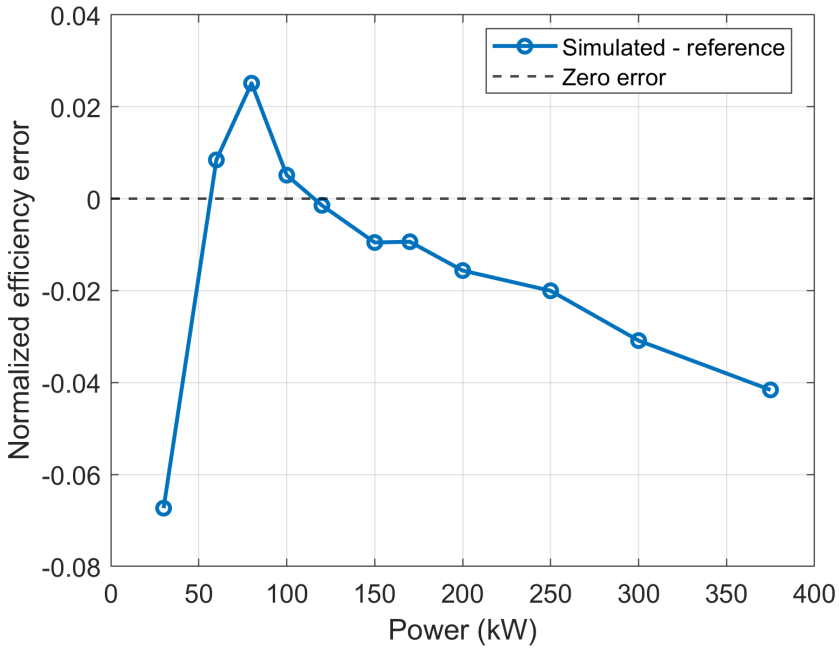


Figure 4.2: Error between the normalized simulated efficiency and the normalized reference efficiency after Bayesian optimization-based parameter identification.

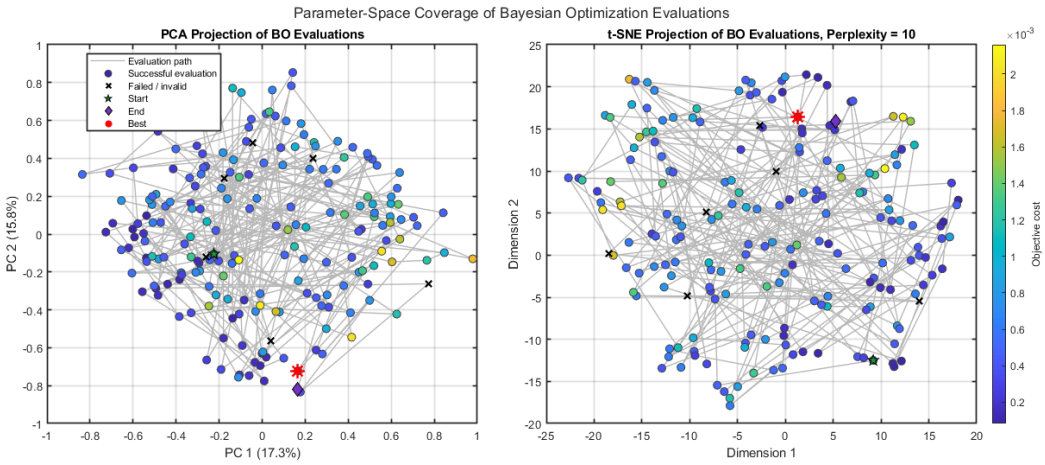


Figure 4.3: Qualitative visualization of parameter-space exploration during Bayesian optimization

Table 4.1: Final calibrated parameters of the PEM fuel cell system model

Parameter	**Symbol**	**Calibrated Value**
Exchange current density	i_0	5.9019×10^{-5}
Limiting current density	i_L	1.5823
Membrane thickness (μm)	t_{mem}	131.7322
Effective cell area (cm^2)	A_{cell}	628.6596
Number of cells	N_{cell}	833
Charge transfer coefficient	α	0.4530
Gas diffusion coefficient (cm^2/s)	D_{GDL}	0.0463
Oxygen stoichiometric ratio	λ_{O_2}	1.7595
Cathode relative humidity	RH_{cat}	0.7475
BoP constant term (kW)	a_0	4.7437
BoP linear coefficient	a_1	0.1496
BoP quadratic coefficient (1/kW)	a_2	1.1856×10^{-4}

4.2 Performance evaluation of machine learning models for internal state predictions

4.2.1 Effect of regression model on prediction accuracy

HER prediction

Table 4.2 summarizes the segment-level K-fold validation results on Group A. SVR achieved the best average validation performance, with the lowest RMSE and the highest R^2. RF and GPR showed comparable average accuracy, but all three models exhibited relatively large standard deviations, indicating that the prediction difficulty varied noticeably among different Group A segments.

Table 4.2: Segment-level K-fold validation results for HER prediction on Group A.

Model	RMSE mean	RMSE std	MAE mean	MAE std	R^2 mean	R^2 std
GPR	0.29818	0.37812	0.09908	0.06752	0.89569	0.18457
SVR	0.25388	0.37736	0.08664	0.07145	0.91310	0.17332
RF	0.28010	0.41010	0.09061	0.07605	0.89588	0.20574

The external validation results on Group B are shown in Table 4.3 and Figure 4.4. The differences among the models become more evident under external validation. RF achieved the best external performance ($R^2 = 0.903$), while GPR showed a clear performance drop despite its competitive K-fold accuracy.

Figure 4.5 further compares the external prediction results using true-versus-predicted scatter plots. The RF and SVR predictions are more concentrated around the diagonal reference line, while GPR contains several large deviations, confirming its weaker generalization on Group B.

Table 4.3: External validation results for HER prediction on Group B.

Model	RMSE	MAE	R^2
GPR	0.65299	0.11089	0.34076
SVR	0.29731	0.06618	0.86333
RF	0.25104	0.07229	0.90256

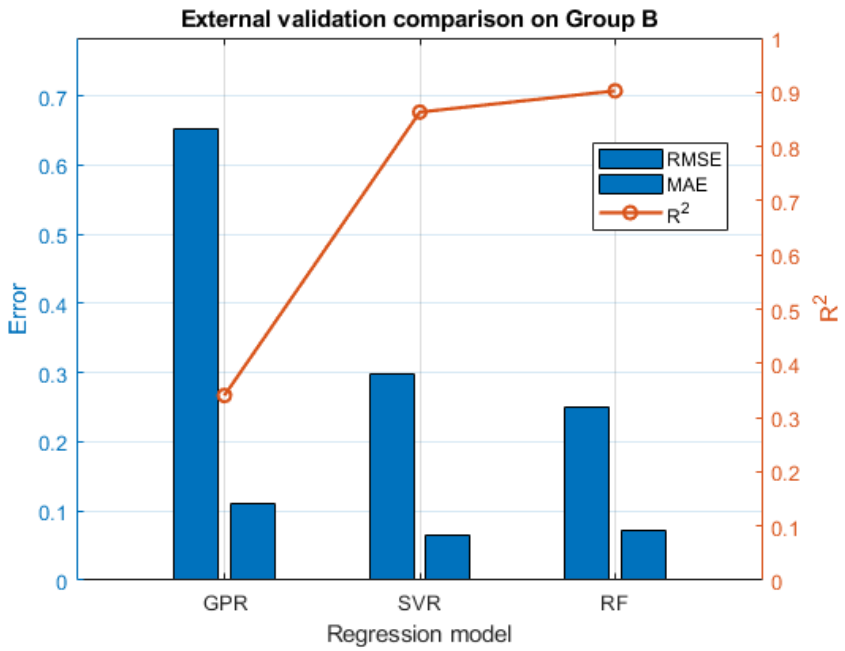


Figure 4.4: External validation comparison of GPR, SVR, and RF for HER prediction on Group B.

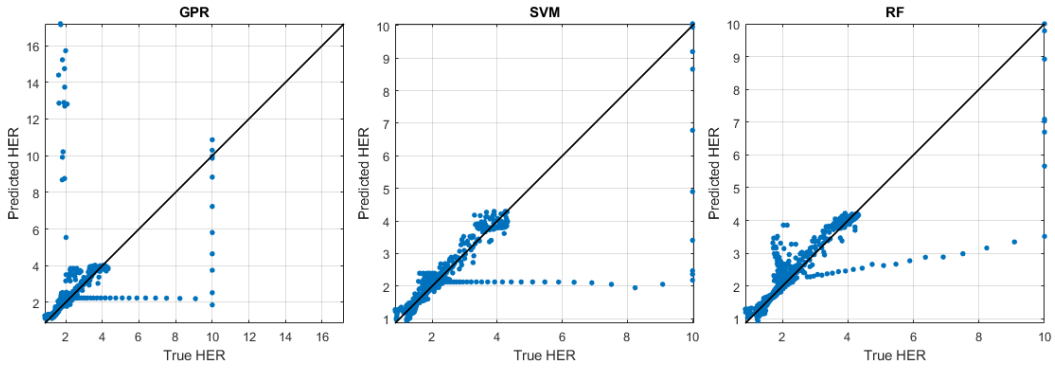


Figure 4.5: External Group B validation scatter plots for HER prediction. The diagonal line represents the ideal prediction $\hat{y} = y$.

Figure 4.6 shows the signed relative error distributions on Group B. GPR exhibits the largest positive median error (1.47%) and the widest IQR, indicating a systematic tendency to overestimate HER. SVR achieves the lowest median error (0.33%) and the smallest IQR, while RF shows an intermediate median of 1.20%.

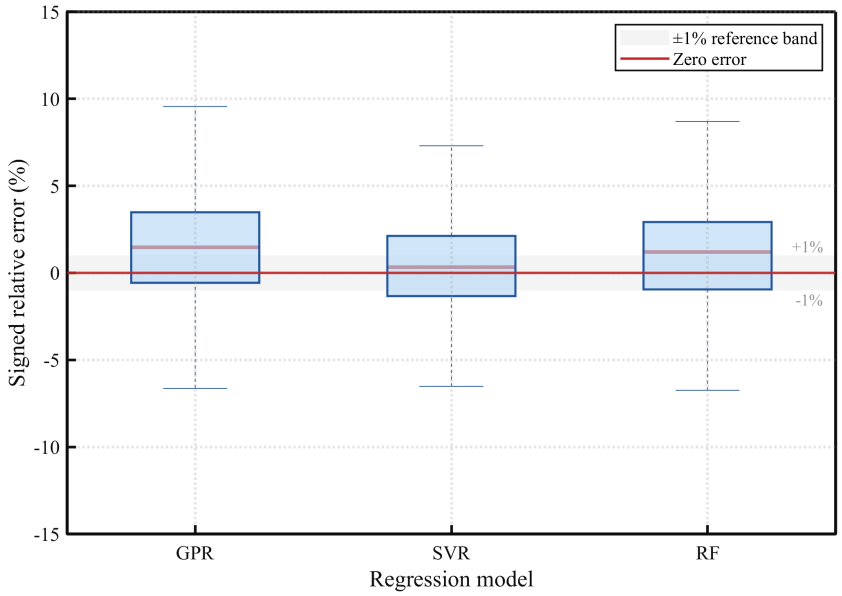


Figure 4.6: Signed relative error distribution of HER prediction on Group B for the three regression models.

The CDF in Figure 4.7 provides a complementary view. SVR achieves the best 90th-percentile threshold (6.41%), followed by RF (7.62%) and GPR (9.42%). This indicates that SVR produces more uniformly accurate percentage predictions, while RF's advantage in RMSE and R^2 reflects better absolute accuracy at larger HER values, which dominate the squared-error metric.

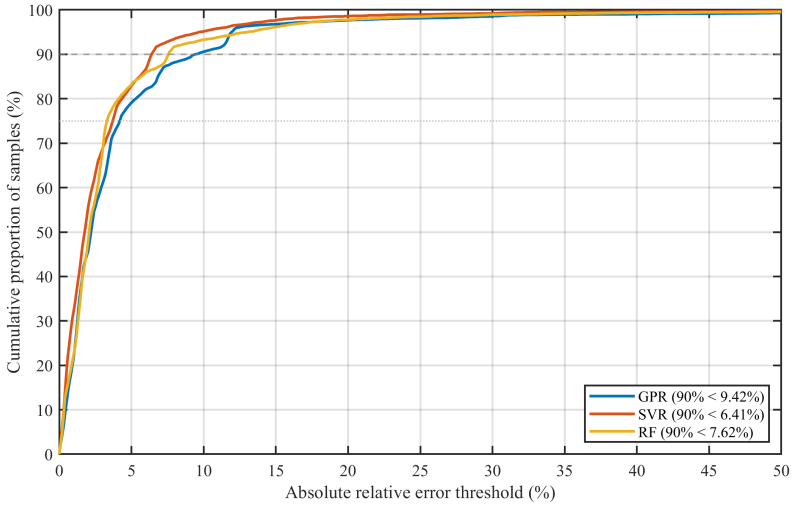


Figure 4.7: Cumulative distribution function of absolute relative error for HER prediction on Group B.

Overall, SVR achieved the best average K-fold performance on Group A and the

most uniform relative error distribution on Group B, while RF achieved the best absolute external validation metrics. Given that external generalization is the primary criterion for the subsequent degradation index, RF is selected as the representative model for HER prediction.

OER prediction

Table 4.4 summarizes the segment-level K-fold validation results for OER prediction on Group A. SVR achieved the best average validation performance, with the lowest RMSE and the highest R^2. GPR and RF also showed acceptable average accuracy, although their validation errors were slightly higher than that of SVR. Similar to the HER case, the relatively large standard deviations indicate that the prediction difficulty varied across different Group A segments.

Table 4.4: Segment-level K-fold validation results for OER prediction on Group A.

Model	RMSE mean	RMSE std	MAE mean	MAE std	R^2 mean	R^2 std
GPR	0.15925	0.28463	0.03173	0.03364	0.93206	0.14711
SVR	0.12485	0.19022	0.03628	0.02357	0.96533	0.07024
RF	0.16229	0.29688	0.02947	0.03399	0.92736	0.15856

The external validation results on Group B are shown in Table 4.5 and Figure 4.8. Compared with HER prediction, all three models achieved high external validation accuracy for OER, with R^2 values above 0.98. RF achieved the best external performance ($R^2 = 0.996$), while GPR also generalized well and SVR showed slightly larger external error.

Table 4.5: External validation results for OER prediction on Group B.

Model	RMSE	MAE	R^2
GPR	0.06276	0.02039	0.98907
SVR	0.08478	0.02873	0.98005
RF	0.03588	0.01449	0.99643

Figure 4.9 further compares the external prediction results using true-versus-predicted scatter plots. All three models show good agreement with the reference OER values, with RF giving the most compact distribution around the ideal prediction line.

Figure 4.10 shows the signed relative error distributions on Group B. All three models show near-zero median errors, but their IQR characteristics differ noticeably. RF achieves the tightest IQR (0.53%), indicating the most consistent prediction across samples. SVR shows the widest IQR (1.63%) despite its near-zero median, suggesting larger variability across segments.

The CDF in Figure 4.11 confirms that RF achieves the best 90th-percentile threshold (2.20%), followed by GPR (2.61%) and SVR (3.51%). Unlike the HER case, where SVR outperformed RF in relative error metrics, RF is here the strongest model across both

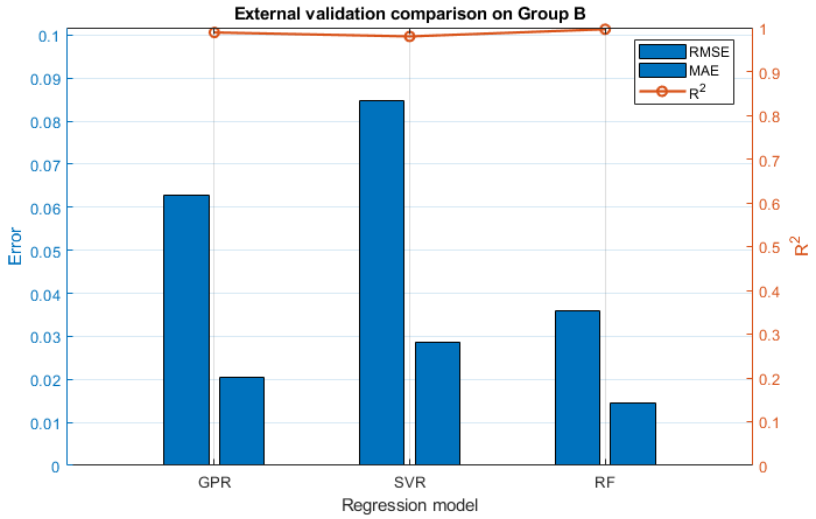


Figure 4.8: External validation comparison of GPR, SVR, and RF for OER prediction on Group B.

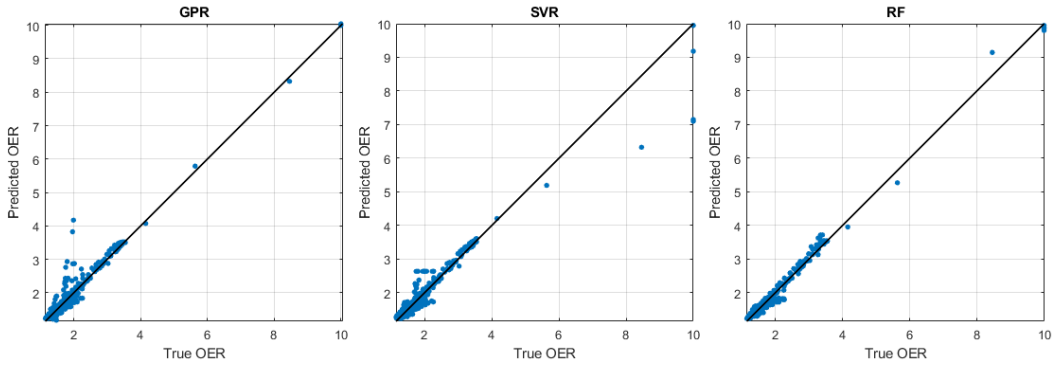


Figure 4.9: External Group B validation scatter plots for OER prediction. The diagonal line represents the ideal prediction $\hat{y} = y$.

absolute and relative criteria simultaneously, providing a consistent advantage over the other two models.

Overall, OER prediction is substantially more accurate than HER across all three models, reflecting its closer dependence on instantaneous operating conditions. RF achieved the best performance on both Group A external validation and relative error distribution, further supporting its selection as the representative regression model.

Membrane water content prediction

Compared with HER and OER, membrane water content is more strongly affected by recent operating history. Therefore, the membrane water content prediction was evaluated using the extended feature set including P_k, $\dot{P}_k$, and 40 s history-dependent load features. The segment-level K-fold validation results on Group A are summarized in Table 4.6. Among the three regression models, RF achieved the best average validation performance,

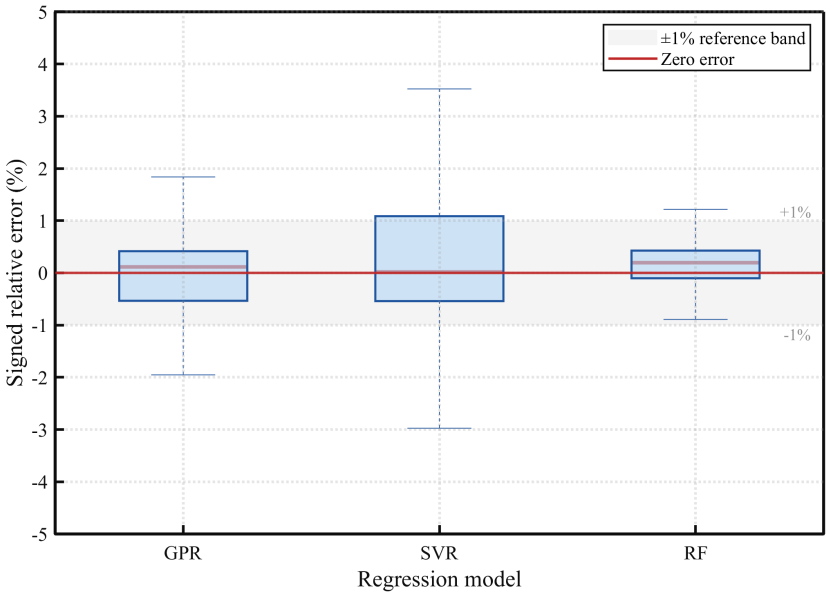


Figure 4.10: Signed relative error distribution of OER prediction on Group B for the three regression models.

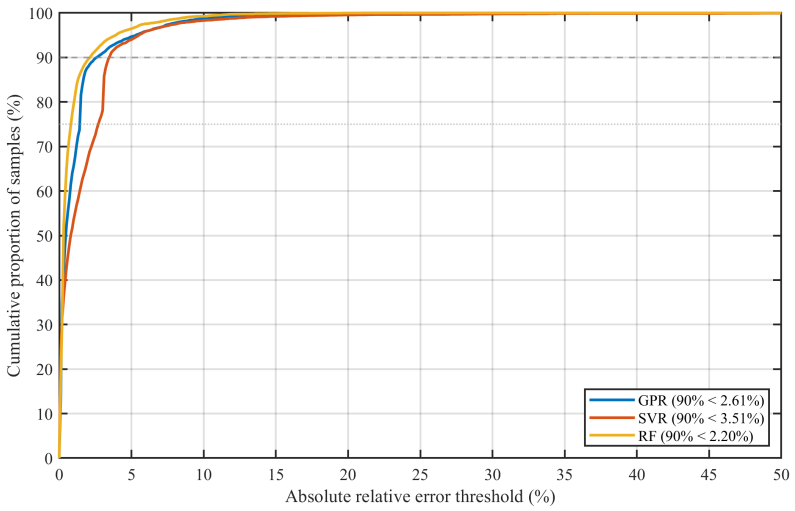


Figure 4.11: Cumulative distribution function of absolute relative error for OER prediction on Group B.

with the lowest RMSE and MAE and the highest mean R^2. GPR showed moderate accuracy, while SVR gave the weakest average validation performance for this target.

The external validation results on Group B are shown in Table 4.7 and Figure 4.12. RF achieved the best external prediction performance ($R^2 = 0.901$), while SVR showed the largest external error despite its competitive K-fold accuracy, indicating weaker generalization to unseen operating profiles.

Table 4.6: Segment-level K-fold validation results for membrane water content prediction on Group A using 40 s history features.

Model	RMSE mean	RMSE std	MAE mean	MAE std	R^2 mean	R^2 std
GPR	0.84655	0.70225	0.34187	0.26795	0.82486	0.18806
SVR	1.02430	0.61233	0.41083	0.18791	0.71969	0.23724
RF	0.64960	0.72250	0.19434	0.22631	0.88418	0.17843

Table 4.7: External validation results for membrane water content prediction on Group B using 40 s history features.

Model	RMSE	MAE	R^2
GPR	0.68996	0.25566	0.74834
SVR	0.80354	0.41501	0.65867
RF	0.43266	0.11248	0.90104

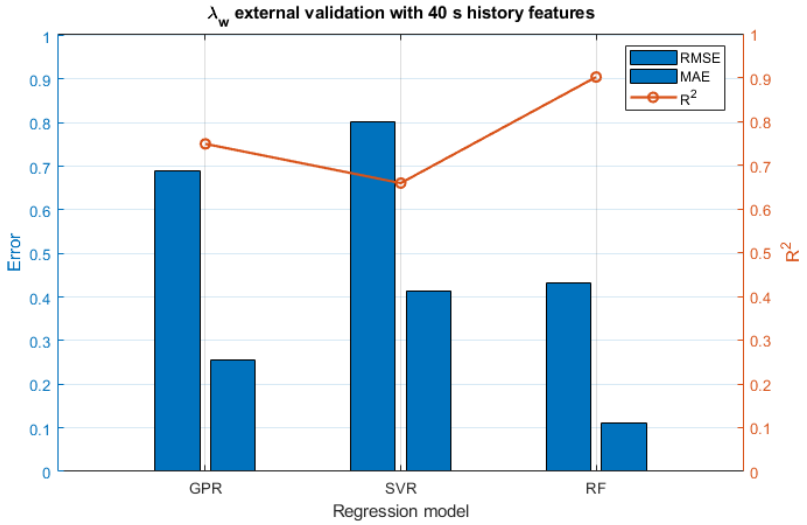


Figure 4.12: External validation comparison of GPR, SVR, and RF for membrane water content prediction on Group B using 40 s history features.

Figure 4.13 further compares the external prediction results using true-versus-predicted scatter plots. The RF predictions are more concentrated around the diagonal reference line, while GPR and SVR show larger deviations especially in regions of stronger membrane water content variation.

The relative error analysis in Figures 4.14 and 4.15 reveals a pattern that contrasts with the absolute metrics. SVR achieves the tightest IQR (0.41%) and the best 90th-percentile threshold (1.32%), while RF shows a heavier error tail and its CDF does not reach 90% coverage within a 20% absolute relative error threshold, yielding a MAPE of 6.13%. This apparent contradiction arises because λ_w is a slowly varying state: many samples cluster near the mean hydration level, where SVR predicts with high relative accuracy. However, SVR fails to capture deviations at extreme hydration values, which drives its poor absolute

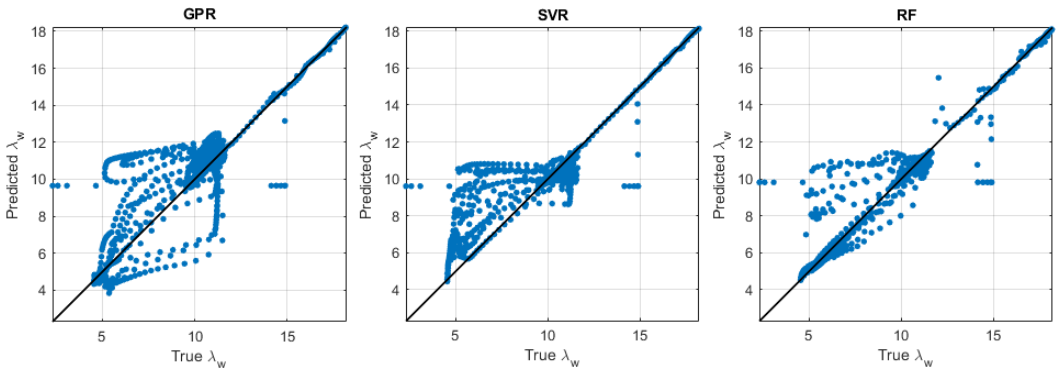


Figure 4.13: External Group B validation scatter plots for membrane water content prediction. The diagonal line represents the ideal prediction $\hat{y} = y$.

metrics. RF, by contrast, tracks these extreme values more accurately in absolute terms but incurs larger relative errors when λ_w takes small values.

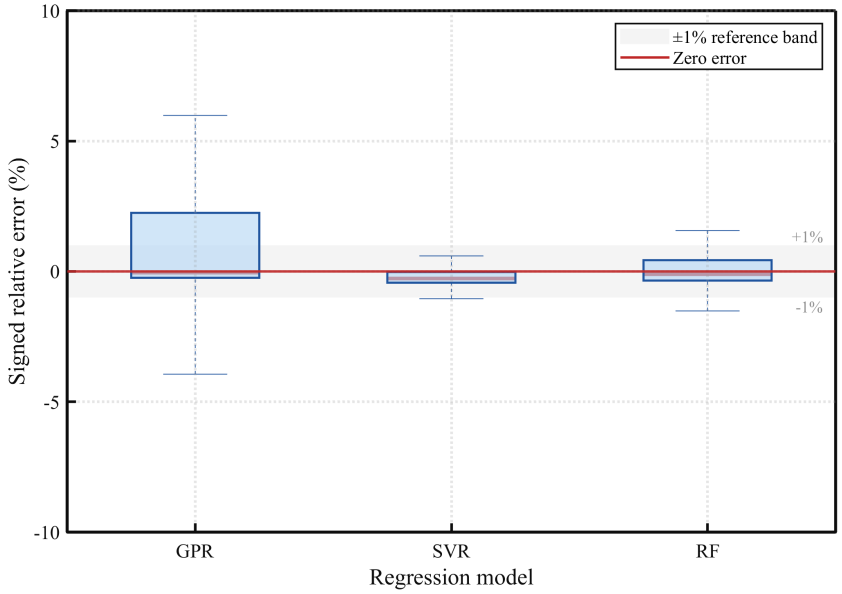


Figure 4.14: Signed relative error distribution of membrane water content prediction on Group B for the three regression models.

Since the degradation severity terms derived from λ_w depend on its absolute value rather than its relative deviation, absolute prediction accuracy is the more relevant criterion here. RF is therefore selected as the representative model for membrane water content prediction, consistent with its superior performance on both K-fold and external absolute

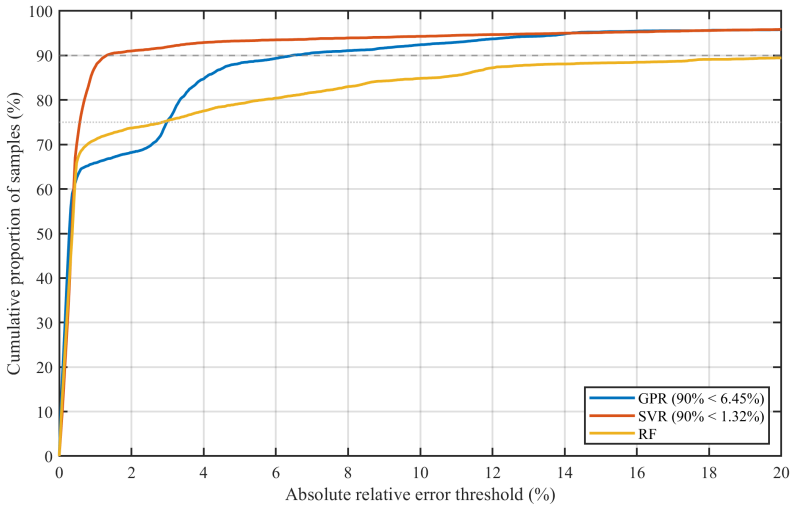


Figure 4.15: Cumulative distribution function of absolute relative error for membrane water content prediction on Group B.

metrics. This also confirms that membrane water content is more history-dependent than HER and OER, and that 40 s load-history descriptors are necessary for robust external generalization.

4.2.2 Effect of feature selection on prediction performance

HER prediction

The influence of input feature selection on HER prediction was evaluated using RF as the fixed regression model, comparing six configurations: P_k only, $P_k + \dot{P}_k$, and four history-dependent feature sets with look-back windows of 5 s, 10 s, 20 s, and 40 s.

Table 4.8 reports the segment-level K-fold validation results on Group A. Adding the power slope slightly improved average performance over P_k only, while the 10 s history window achieved the best mean RMSE among all configurations. However, the improvements are moderate and accompanied by large standard deviations, indicating considerable variation in prediction difficulty across segments.

Table 4.8: Segment-level K-fold validation results for HER prediction using RF with different feature configurations.

Feature set	RMSE mean	RMSE std	MAE mean	MAE std	R^2 mean	R^2 std
P_k only	0.29545	0.39692	0.09741	0.07522	0.89285	0.19993
$P_k + \dot{P}_k$	0.27984	0.41043	0.09059	0.07607	0.89583	0.20595
History 5 s	0.26961	0.41778	0.08461	0.07711	0.89786	0.20848
History 10 s	0.26092	0.41947	0.08259	0.07835	0.90012	0.20723
History 20 s	0.29189	0.40765	0.08881	0.07888	0.88368	0.20230
History 40 s	0.30251	0.40675	0.09107	0.07807	0.87378	0.20274

The external validation results on Group B, summarized in Table 4.9 and Figure 4.16, reveal a clearer trend. Adding $\dot{P}_k$ to P_k alone improves the external R^2 from 0.875 to 0.905. History-dependent features provide further improvement, with the 40 s window achieving the best external result ($R^2 = 0.986$), indicating that recent load history improves generalization to unseen profiles.

Table 4.9: External validation results for HER prediction using RF with different feature configurations.

Feature set	RMSE	MAE	R^2
P_k only	0.28507	0.07813	0.87435
$P_k + \dot{P}_k$	0.24841	0.07162	0.90460
History 5 s	0.14953	0.06761	0.96543
History 10 s	0.13147	0.07633	0.97328
History 20 s	0.13193	0.08120	0.97309
History 40 s	0.09629	0.05574	0.98567

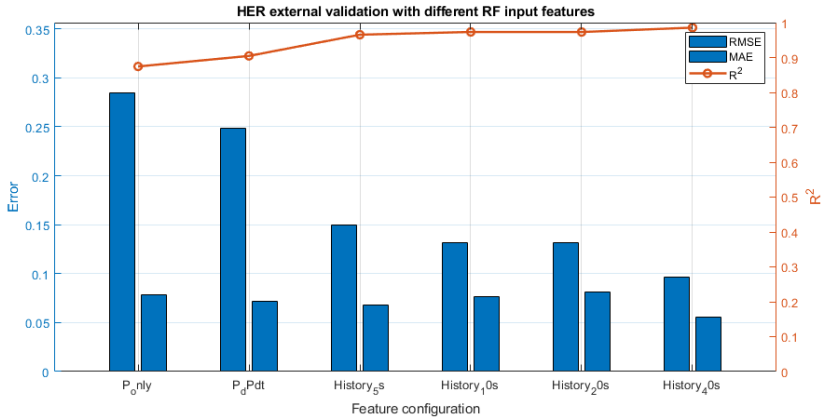


Figure 4.16: External validation comparison of RF-based HER prediction using different input feature configurations.

Figure 4.17 shows that all configurations share a small positive median error, reflecting a systematic tendency to overestimate HER during rapid load transients. The interquartile ranges remain similar across configurations, suggesting that the RMSE improvement brought by history features occurs primarily in the tail of the error distribution rather than in the bulk of predictions.

This is confirmed by the CDF analysis in Figure 4.18. History 10 s and History 20 s exhibit a flat region between approximately 5% and 15% absolute relative error, indicating a bimodal error distribution where most samples are predicted well but a subset incurs larger errors. By contrast, $P_k + \dot{P}_k$ and History 40 s both show smooth, unimodal CDFs with similar 90th-percentile thresholds of 7.7% and 6.7%, respectively.

The segment-wise time-series comparison in Figure 4.19 further illustrates these differences. The P_k-only model produces flatter responses during transients, while adding $\dot{P}_k$ substantially improves transient tracking. Although history-based configurations

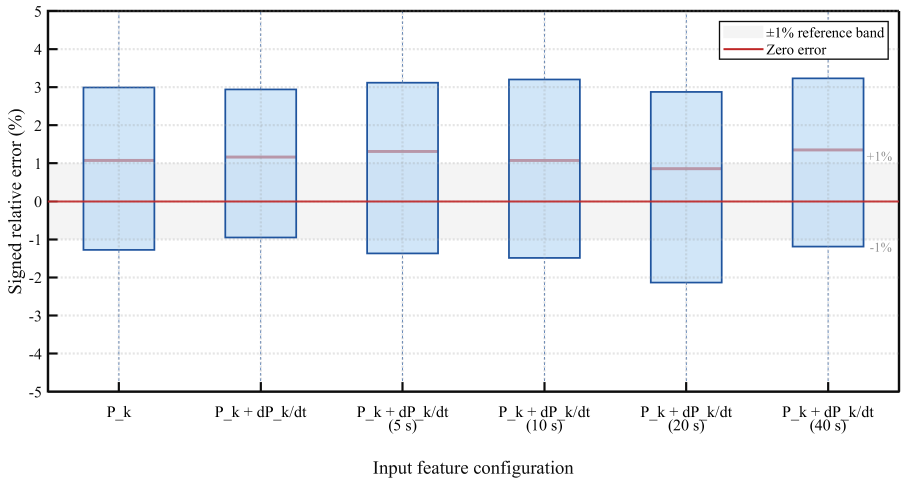


Figure 4.17: Signed relative error distribution of HER prediction on Group B across six input feature configurations.

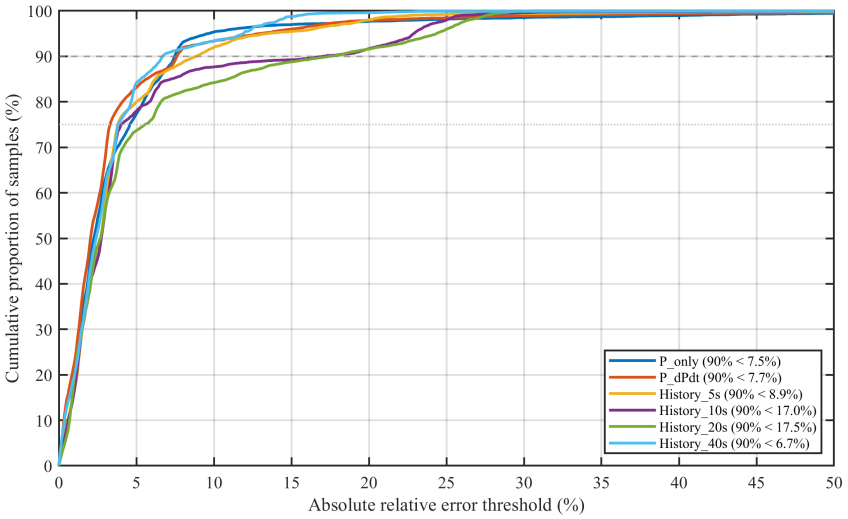


Figure 4.18: Cumulative distribution function of absolute relative error for HER prediction on Group B.

achieve lower overall external errors, they can introduce local non-smooth artefacts in some transient and recovery regions, such as step-like deviations and short oscillations, which are likely related to the piecewise nature of the RF model and the higher-dimensional window-based statistics.

Based on these results, $P_k + \dot{P}_k$ is selected as the preferred input configuration for HER prediction. It achieves good external generalization ($R^2 = 0.905$) with a unimodal error distribution, avoids the local artefacts observed for History 10 s and History 20 s, and offers only a marginal CDF gap relative to History 40 s while requiring no sliding history buffer. History-based configurations are therefore treated as accuracy-oriented extensions

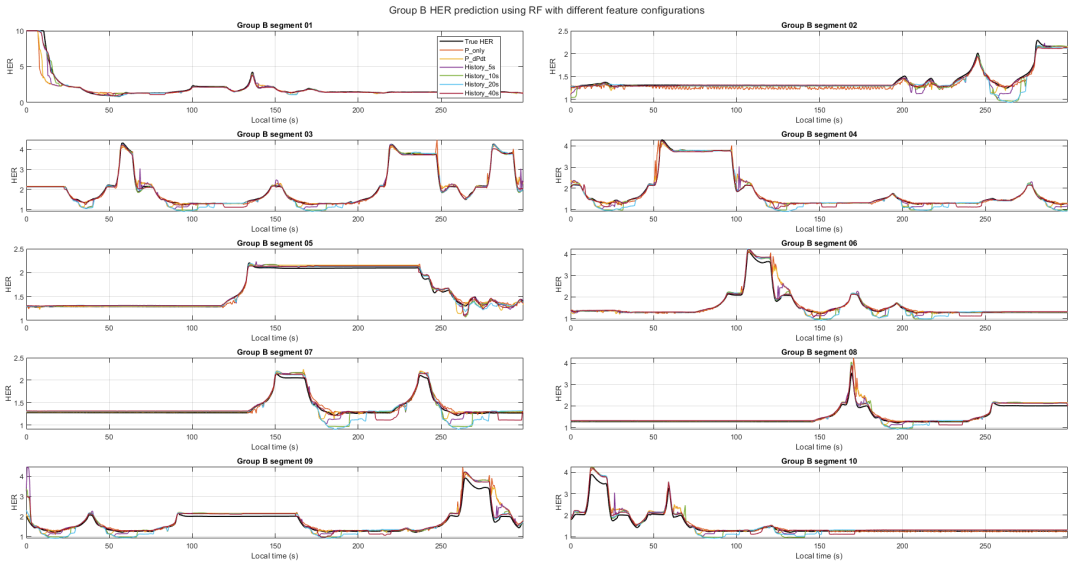


Figure 4.19: Segment-wise Group B HER prediction using RF with different input feature configurations.

rather than the preferred compact estimator.

OER prediction

The same six feature configurations were evaluated for OER prediction using RF as the fixed regression model.

Table 4.10 reports the segment-level K-fold validation results on Group A. Adding $\dot{P}_k$ substantially improved average performance over P_k only, whereas the history-based feature sets provided only marginal further improvement. This suggests that most of the useful dynamic information for OER prediction is already captured by the instantaneous power and power slope.

Table 4.10: Segment-level K-fold validation results for OER prediction using RF with different feature configurations.

Feature set	RMSE mean	RMSE std	MAE mean	MAE std	R^2 mean	R^2 std
P_k only	0.20533	0.27679	0.06277	0.03884	0.91633	0.15660
$P_k + \dot{P}_k$	0.16190	0.29711	0.02944	0.03404	0.92739	0.15858
History 5 s	0.16010	0.29823	0.02748	0.03460	0.92754	0.15878
History 10 s	0.16062	0.29795	0.02744	0.03435	0.92741	0.15871
History 20 s	0.16227	0.29697	0.02868	0.03446	0.92722	0.15855
History 40 s	0.16237	0.29704	0.02979	0.03420	0.92724	0.15871

The external validation results, summarized in Table 4.11 and Figure 4.20, confirm this pattern more clearly. Adding $\dot{P}_k$ reduces the external RMSE from 0.100 to 0.036 and raises R^2 to 0.996. History-dependent features achieve marginally lower RMSE only for the 5 s window, while longer windows of 20 s and 40 s slightly degrade external performance, indicating no practical advantage over $P_k + \dot{P}_k$ for OER prediction.

Table 4.11: External validation results for OER prediction using RF with different feature configurations.

Feature set	RMSE	MAE	R^2
P_k only	0.10008	0.05500	0.97220
$P_k + \dot{P}_k$	0.03624	0.01446	0.99636
History 5 s	0.03455	0.01274	0.99669
History 10 s	0.03602	0.01264	0.99640
History 20 s	0.03941	0.01562	0.99569
History 40 s	0.04005	0.01660	0.99555

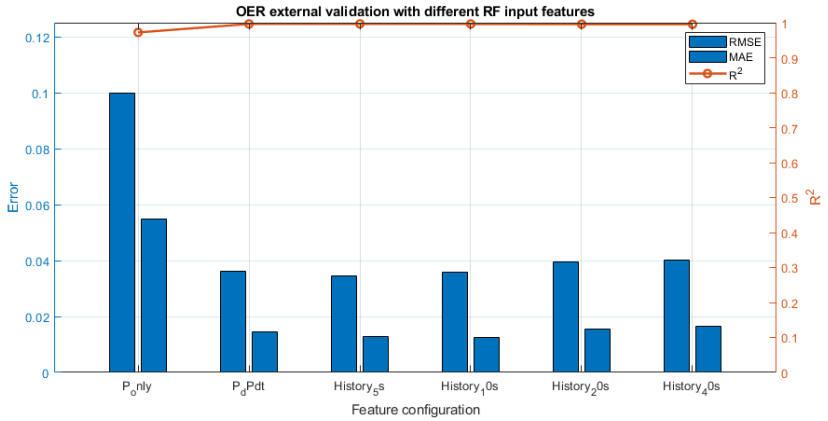


Figure 4.20: External validation comparison of RF-based OER prediction using different input feature configurations.

Figure 4.21 shows that adding $\dot{P}_k$ narrows the IQR from approximately $\pm 1\%$ to within the $\pm 1\%$ reference band, and the median error remains close to zero across all configurations. Further extending the feature set with history windows produces no visible distributional improvement.

The CDF in Figure 4.22 confirms this. The P_k-only curve is clearly separated from the remaining five, with a 90th-percentile threshold of 11.7% compared with 2.2% for $P_k + \dot{P}_k$. All configurations including $\dot{P}_k$ show smooth, unimodal CDFs, and the improvement from $P_k + \dot{P}_k$ to the best-performing History 10 s is limited to 0.4 percentage points. The 20 s and 40 s windows perform slightly worse than $P_k + \dot{P}_k$, suggesting that longer history introduces minor generalisation degradation for OER.

The segment-wise comparison in Figure 4.23 further supports this conclusion. Adding $\dot{P}_k$ substantially improves transient tracking over P_k only, while history-based configurations offer only marginal local improvements and occasionally introduce small non-smooth responses.

Based on these results, $P_k + \dot{P}_k$ is selected as the preferred feature configuration for OER prediction. OER dynamics are primarily governed by the instantaneous operating point and the current load-change rate, and extended load-history descriptors provide no meaningful benefit.

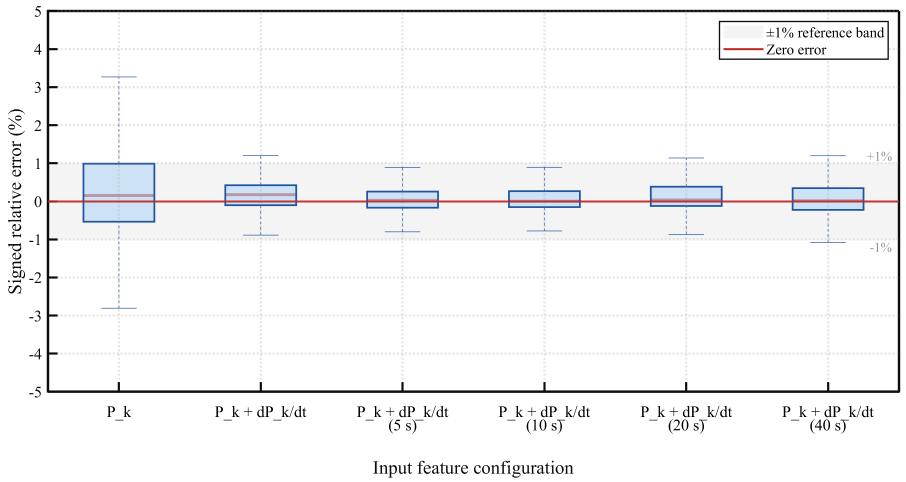


Figure 4.21: Signed relative error distribution of OER prediction on Group B across six input feature configurations.

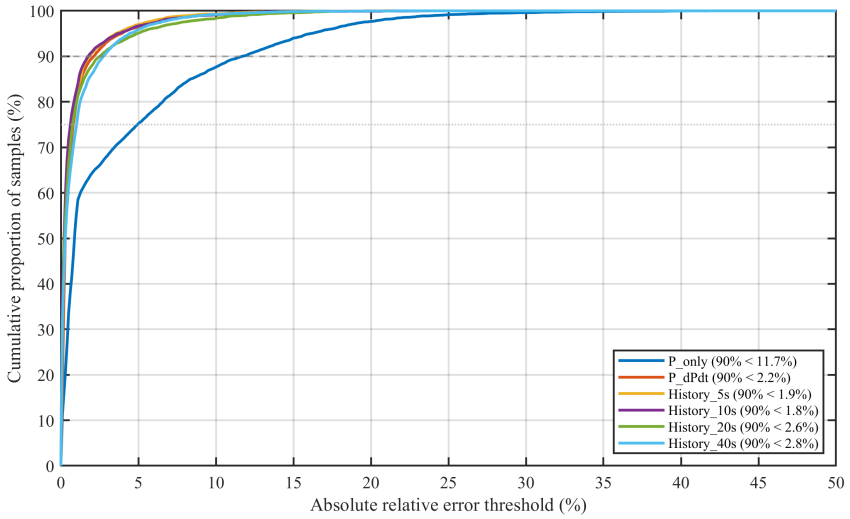


Figure 4.22: Cumulative distribution function of absolute relative error for OER prediction on Group B.

Membrane water content prediction

The same six feature configurations were evaluated for membrane water content prediction using RF as the fixed regression model. Unlike HER and OER, λ_w is expected to have stronger dynamic memory due to the slower processes of water generation, membrane hydration, humidification, and transport.

Table 4.12 reports the segment-level K-fold validation results on Group A. Adding $\dot{P}_k$ improved average performance over P_k only, and the 5 s and 10 s history windows achieved the lowest mean RMSE values. The 40 s window obtained the lowest mean MAE. The large standard deviations across all configurations reflect the strongly segment-dependent

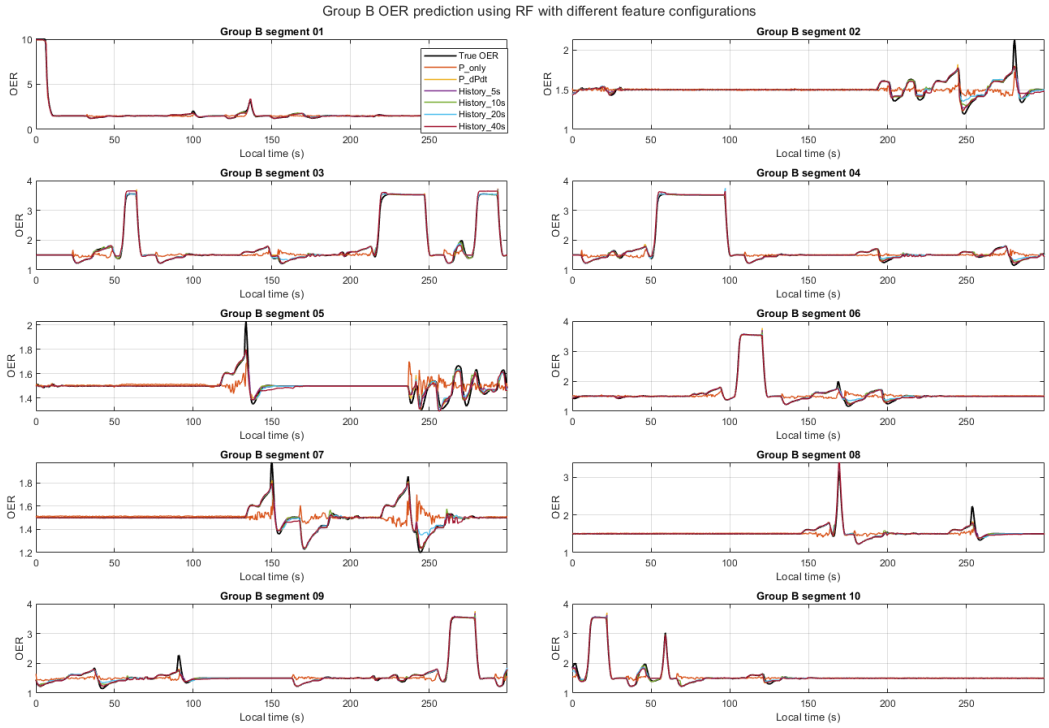


Figure 4.23: Segment-wise Group B OER prediction using RF with different input feature configurations.

nature of membrane hydration dynamics.

Table 4.12: Segment-level K-fold validation results for membrane water content prediction using RF with different feature configurations.

Feature set	RMSE mean	RMSE std	MAE mean	MAE std	R^2 mean	R^2 std
P_k only	0.93305	0.55588	0.33980	0.21375	0.81203	0.13630
$P_k + \dot{P}_k$	0.74635	0.65145	0.27080	0.22782	0.86625	0.16874
History 5 s	0.61001	0.71325	0.21467	0.22179	0.89355	0.17942
History 10 s	0.60269	0.72765	0.20418	0.21201	0.89013	0.18195
History 20 s	0.76210	0.68304	0.24168	0.21504	0.83786	0.18296
History 40 s	0.65061	0.72215	0.19464	0.22719	0.88419	0.17925

The external validation results on Group B, summarized in Table 4.13 and Figure 4.24, reveal a striking contrast with HER and OER. Both P_k only and $P_k + \dot{P}_k$ yield negative external R^2 values, and short history windows of 5 s, 10 s, and 20 s fail to recover reliable generalization. Only the 40 s history window achieves good external performance, with $R^2 = 0.900$, demonstrating that λ_w is governed by a longer recent operating history that cannot be captured by instantaneous or short-window features.

Figure 4.25 shows that the IQR and median error remain similar across all six configurations, with most samples falling within the $\pm 1\%$ reference band. However, since λ_w is a slowly varying state, many samples stay near the normal hydration level, making the IQR an insensitive discriminator between feature sets. The CDF analysis is therefore

Table 4.13: External validation results for membrane water content prediction using RF with different feature configurations.

Feature set	RMSE	MAE	R^2
P_k only	1.69427	0.63346	-0.51751
$P_k + \dot{P}_k$	1.65512	0.64937	-0.44819
History 5 s	1.50268	0.60008	-0.19370
History 10 s	1.38101	0.57344	-0.00823
History 20 s	1.46942	0.58234	-0.14145
History 40 s	0.43425	0.11157	0.90031

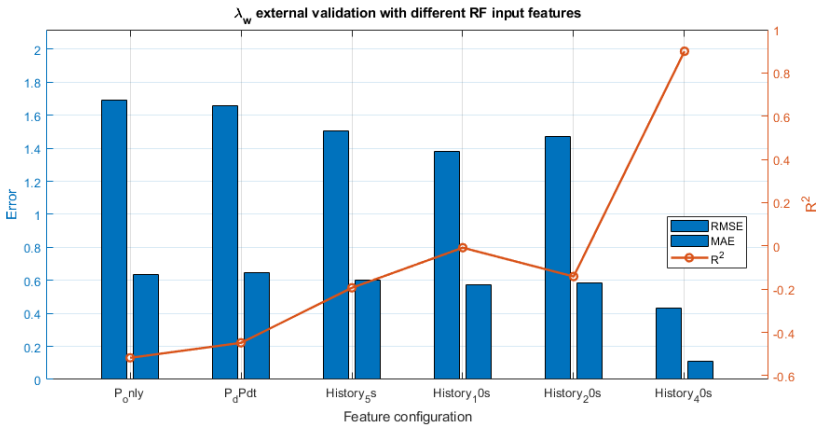


Figure 4.24: External validation comparison of RF-based membrane water content prediction using different input feature configurations.

more informative for comparing large-error behaviour.

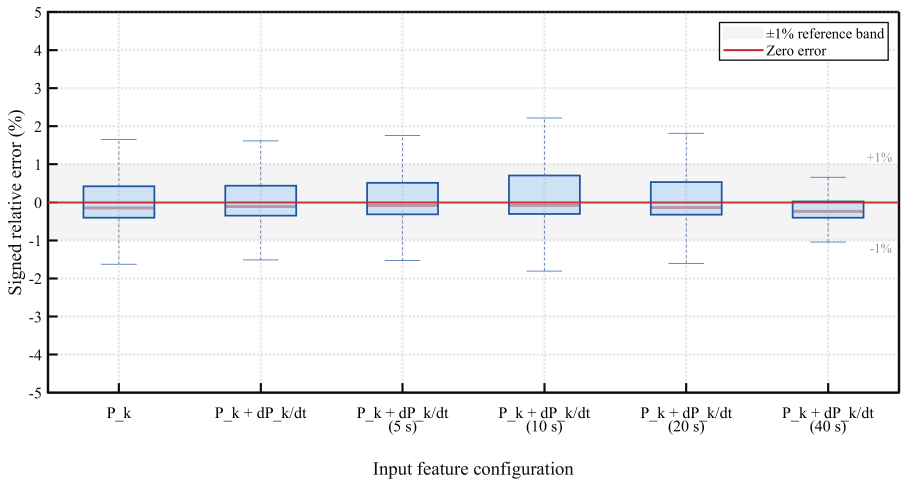


Figure 4.25: Signed relative error distribution of λ_w prediction on Group B across six input feature configurations.

Figure 4.26 shows a clear separation between History 40 s and all other configurations. The five shorter feature sets have 90th-percentile error thresholds between 21.6% and 24.5% and their CDF curves are nearly indistinguishable, indicating that neither the power slope nor short history windows add meaningful information for λ_w on Group B. By contrast, the 40 s window reduces the 90th-percentile threshold to 1.7%, with its CDF rising steeply in the low-error region.

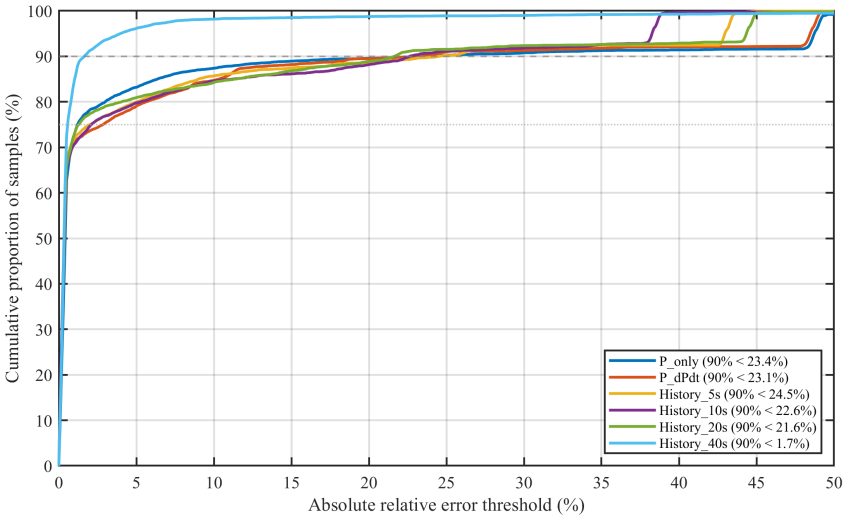


Figure 4.26: Cumulative distribution function of absolute relative error for λ_w prediction on Group B.

The segment-wise comparison in Figure 4.27 provides a qualitative explanation. The compact models and short history windows fail to track sustained high or low hydration states, as they cannot distinguish whether the current operating point follows a prolonged high-load or low-load history. The 5–20 s configurations also produce artificial plateaus or delayed recovery in several segments. The 40 s history window follows the main λ_w trend consistently across Group B, confirming that a longer look-back window is necessary to capture the relevant membrane hydration dynamics.

Based on these results, the 40 s history feature set is selected as the preferred configuration for λ_w prediction. This contrasts with HER and OER, where $P_k + \dot{P}_k$ was sufficient, and reflects the fundamentally slower dynamics of membrane hydration.

4.2.3 Monte Carlo noise sensitivity analysis

To evaluate the sensitivity of the estimators to power measurement uncertainty, Gaussian noise at three levels ($\sigma \in \{1\%, 2\%, 5\%\}$ of P_{max}) was added to the Group B power signal before feature extraction, and each noise level was repeated 50 times to obtain stable statistics. The results are shown in Figure 4.28 and Table 4.14.

The three estimators respond differently to input noise. HER prediction degrades most severely in absolute terms: at $\sigma = 5\%$, the R^2 falls from 0.905 to 0.595 and the RMSE more than doubles. This is expected, since the $P_k + \dot{P}_k$ feature set directly uses the

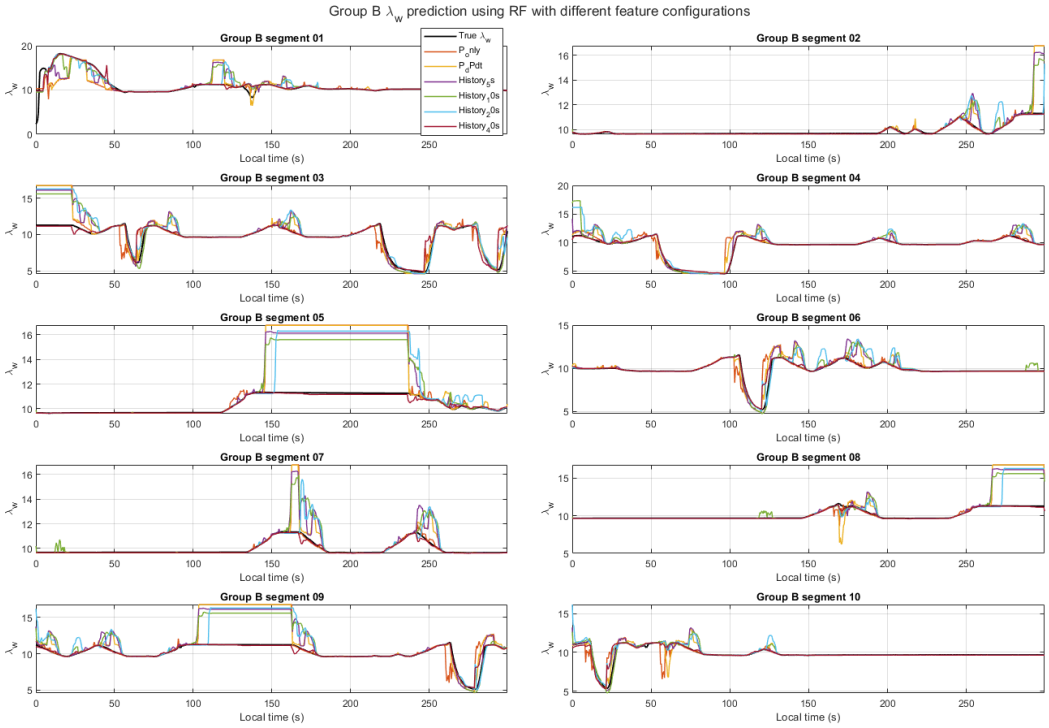


Figure 4.27: Segment-wise Group B membrane water content prediction using RF with different input feature configurations.

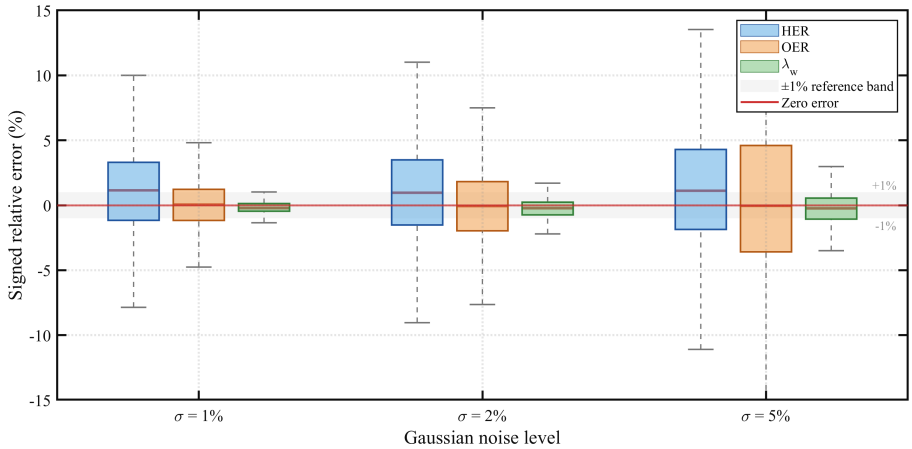


Figure 4.28: Signed relative error distributions of HER, OER, and λ_w prediction under Gaussian power measurement noise at three levels.

instantaneous power slope, which is obtained by differencing consecutive power samples and is therefore highly sensitive to sample-level noise. The systematic positive median error of approximately $+1\%$ persists across all noise levels, confirming that it originates from the model structure rather than measurement quality.

OER shows the largest percentage RMSE increase among the three targets (up to $+176\%$ at $\sigma = 5\%$), but the absolute prediction quality remains high: the R^2 stays above

Table 4.14: RMSE and R^2 of the three estimators under Gaussian noise. Values are reported as mean $\pm$ std over 50 Monte Carlo trials. RMSE increase is relative to the clean baseline.

Target	σ	Baseline R^2	RMSE (noisy)	R^2 (noisy)	RMSE increase
HER	1%	0.905	0.318 ± 0.008	0.844 ± 0.008	$+27.9\%$
	2%		0.355 ± 0.015	0.805 ± 0.016	$+42.8\%$
	5%		0.512 ± 0.014	0.595 ± 0.022	$+105.9\%$
OER	1%	0.996	0.052 ± 0.001	0.992 ± 0.000	$+44.2\%$
	2%		0.067 ± 0.001	0.987 ± 0.000	$+85.4\%$
	5%		0.100 ± 0.001	0.972 ± 0.001	$+175.8\%$
λ_w	1%	0.900	0.633 ± 0.009	0.788 ± 0.006	$+45.7\%$
	2%		0.669 ± 0.010	0.763 ± 0.007	$+54.0\%$
	5%		0.791 ± 0.015	0.669 ± 0.013	$+82.1\%$

0.972 even at the highest noise level. This is because the baseline RMSE of OER is very small (0.036), so the absolute degradation is limited even when the percentage increase is large. The boxplot in Figure 4.28 confirms this: the OER error distribution expands substantially at $\sigma = 5\%$, but the median remains close to zero throughout.

Membrane water content shows a moderate and relatively stable degradation pattern. The RMSE increases by 46% at $\sigma = 1\%$ but only by 82% at $\sigma = 5\%$, a slower rate than the other two targets. The 40 s history window introduces a natural smoothing effect, as the window-based statistical features (mean, standard deviation, slope ratios) average out high-frequency noise in the power signal, attenuating the impact of individual noisy samples. Correspondingly, the boxplot shows that the λ_w error distribution remains relatively compact across noise levels compared with HER and OER.

4.2.4 Robustness evaluation under unmodelled operating conditions

Two parameter sweeps were conducted to assess the sensitivity of the trained estimators to operating conditions not included in the input feature set: a cathode humidity sweep (RH$\in \{0.65, 0.75, 0.85\}$ at fixed $T = 25$ °C) and a temperature sweep ($T \in \{15, 25, 35\}$ °C at fixed $\mathrm{RH}_{\mathrm{cat}} = 0.78$). The resulting R^2 values are summarised in Table 4.15.

Figure 4.29 shows the signed relative error distributions under the humidity sweep. OER prediction is unaffected across all three humidity levels, with R^2 remaining at 0.998 and the error distribution showing no visible shift. HER prediction is similarly insensitive to humidity variation: the R^2 values range from 0.913 to 0.936, and the systematic positive median error of approximately $+1.3\%$ identified in Section 4.2.2 persists unchanged. Membrane water content shows a progressive negative shift in the median error as RH increases, from -0.36% at RH = 0.65 to -1.01% at RH = 0.85, while the R^2 falls from 0.463 to 0.241. This reflects a systematic underestimation of λ_w under higher humidity conditions: a higher cathode humidity increases the actual membrane water content

Table 4.15: R^2 values of the three estimators under humidity and temperature variation.

Sweep	Condition	HER	OER	λ_w
Humidity	RH = 0.65	0.936	0.998	0.463
	RH = 0.75	0.913	0.998	0.407
	RH = 0.85	0.918	0.998	0.241
Temperature	$T = 15$ °C	0.916	0.998	0.537
	$T = 25$ °C	0.913	0.998	0.407
	$T = 35$ °C	0.869	0.996	−0.094

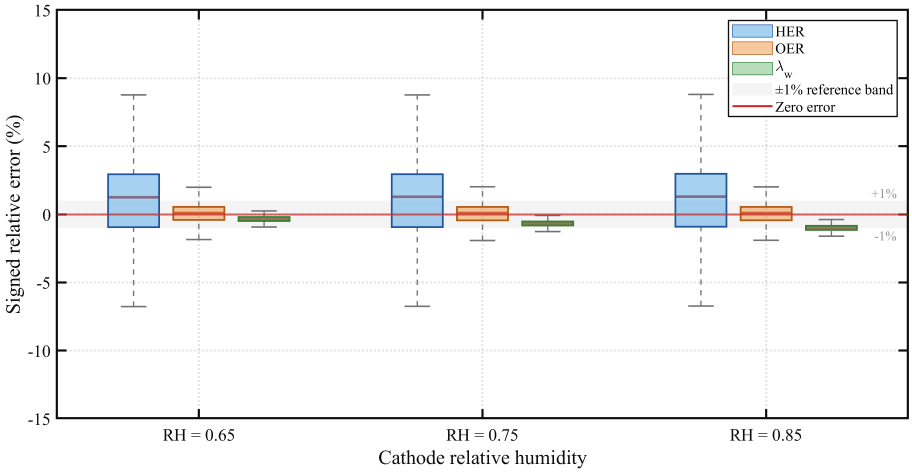


Figure 4.29: Signed relative error distributions under cathode humidity variation at $T = 25$ °C.

relative to the training distribution, but this effect is invisible to the estimator since RH is not included as an input feature.

Figure 4.30 shows the results under the temperature sweep. OER remains robust, with R^2 above 0.996 at all temperatures. For HER, the positive median error increases slightly from $+1.16\%$ at 15°C to $+1.40\%$ at 35°C, and the RMSE rises from 0.314 to 0.409, indicating a moderate sensitivity to elevated temperature. Membrane water content is the most sensitive to temperature variation: the median error shifts from $+0.11\%$ at 15°C to -1.48% at 35°C, and the R^2 declines from 0.537 to -0.094. The negative R^2 at 35°C indicates that the temperature-induced shift in membrane water dynamics falls outside the training distribution and cannot be compensated by the load-history features.

Overall, OER prediction is robust to both temperature and humidity variation. HER maintains acceptable accuracy across all conditions, with a moderate reduction at elevated temperature. Membrane water content is the most sensitive variable: both shifts produce a systematic negative bias that grows with the deviation from the training condition, reflecting the strong dependence of membrane hydration on variables absent from the current input feature set.

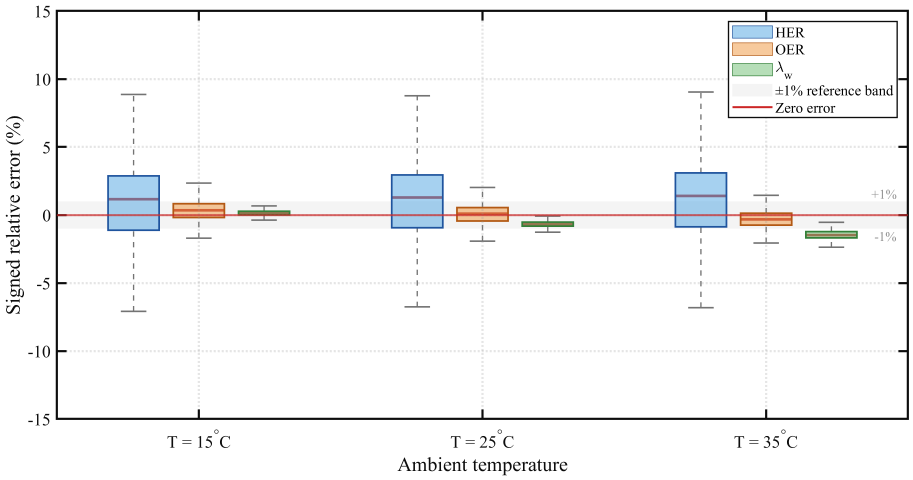


Figure 4.30: Signed relative error distributions under temperature variation at $\mathrm{RH}_{\mathrm{cat}} = 0.78$.

4.2.5 Conclusion on machine learning model

The feature-selection results indicate that different internal states require different input representations. For HER and OER, the compact feature set $P_k + \dot{P}_k$ was selected as the preferred input representation. Figures 4.31 and 4.32 show the learned RF mappings from fuel-cell power and power slope to HER and OER, respectively. Both surfaces exhibit a clear nonlinear dependence on the operating power, while the power slope provides additional information in transient regions. This supports the conclusion that HER and OER can be effectively estimated using instantaneous load information without requiring extended load-history descriptors.

In contrast, membrane water content showed a much stronger dependence on recent operating history. The compact feature set $P_k + \dot{P}_k$ failed to generalize reliably to Group B for λ_w, while the 40 s history-based feature set significantly improved the external validation performance. Figure 4.33 shows the top five predictor importance values for λ_w prediction using the 40 s history feature set. The most important feature is the current power, but history-dependent descriptors, such as the mean power and low-power ratio over the recent window, also contribute to the prediction. This confirms that λ_w cannot be represented as a purely instantaneous mapping from current load conditions, but instead requires information about the recent load trajectory.

Overall, the results suggest a physically consistent feature-selection strategy. The gas supply indicators HER and OER are mainly governed by the current operating point and the instantaneous load-changing trend, so $P_k + \dot{P}_k$ provides a compact, interpretable, and computationally efficient representation. In contrast, membrane water content has slower dynamics and stronger memory effects, so the 40 s history-based feature set is required to achieve reliable external prediction. Therefore, the final feature configurations are selected as $P_k + \dot{P}_k$ for HER and OER prediction, and $P_k + \dot{P}_k$ with 40 s history descriptors for λ_w prediction.

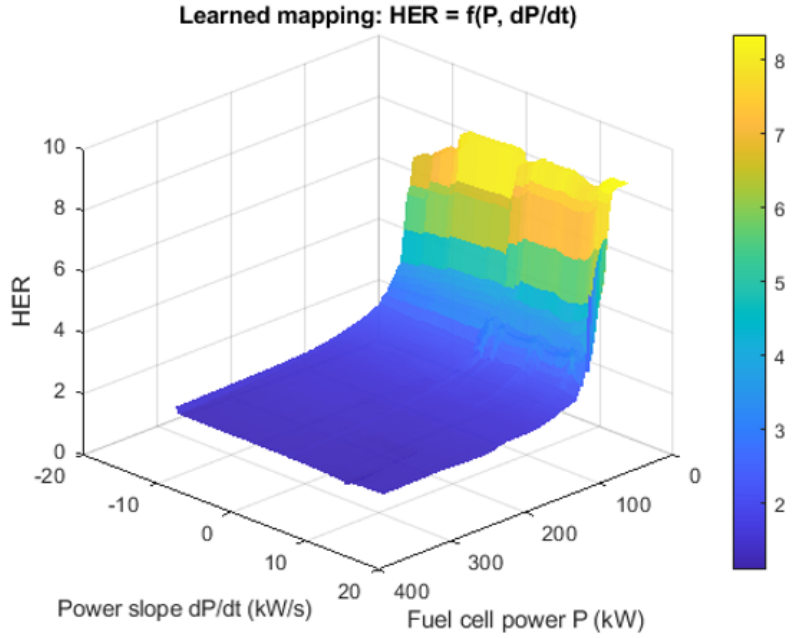


Figure 4.31: Learned RF mapping for HER prediction using the compact feature set $P_k + \dot{P}_k$.

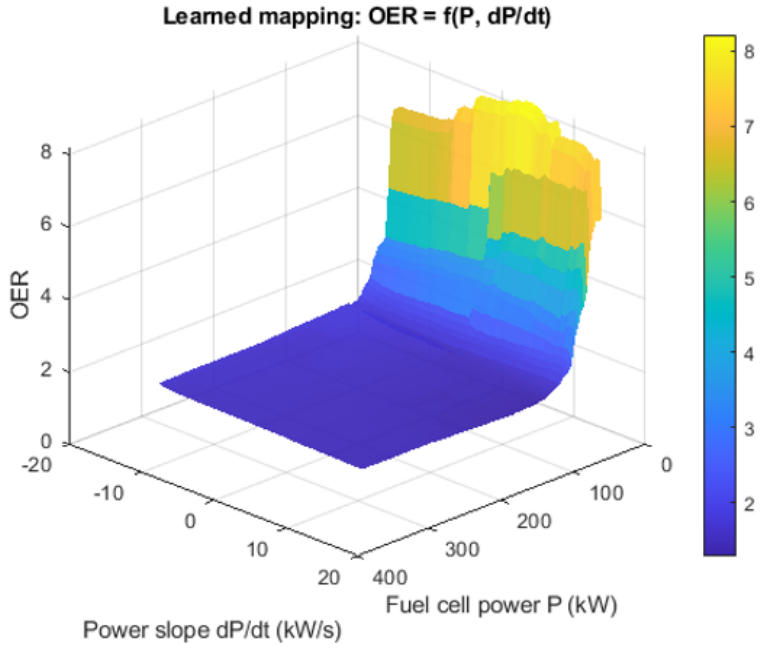


Figure 4.32: Learned RF mapping for OER prediction using the compact feature set $P_k + \dot{P}_k$.

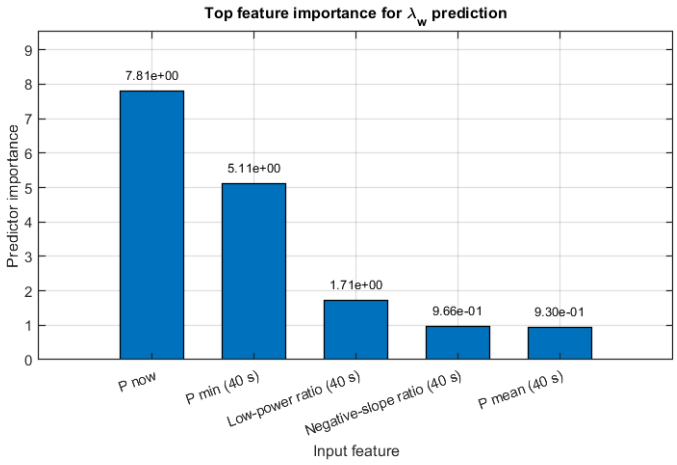


Figure 4.33: Top five RF predictor importance values for membrane water content prediction using the 40 s history-based feature set.

4.3 Validation of the final Degradation Index

4.3.1 Calibrated degradation-index parameters and learned contributions

The final stage of the proposed framework is to validate whether the grey-box degradation index can reproduce the experimentally measured voltage loss rates. After the internal-state estimators have been selected, the degradation-index parameters are calibrated using the reference degradation dataset described in Section 3.4.1. The calibrated reference values, threshold, and weighting coefficients are summarized in Table 4.16, and the learned weight distribution is shown in Fig. 4.34.

Table 4.16: Calibrated parameters and weights of the grey-box degradation index.

Category	Symbol	Value
HER reference	$\lambda_{H_2}^{\text{ref}}$	1.7347
OER reference	$\lambda_{O_2}^{\text{ref}}$	1.5119
Membrane water reference	λ_{w}^{ref}	13.3165
High-power threshold	$P_{\text{high}}^{\text{ref}}$	180.0058 kW
HER weight	w_H	0.0018
OER weight	w_O	0.2384
Membrane state weight	$w_{m,\text{state}}$	0.0349
Membrane cycling weight	$w_{m,\text{cycle}}$	0.2923
High-power weight	w_{highP}	0.0006
Power cycling weight	$w_{P,\text{cycle}}$	0.2712
Power amplitude weight	w_{amp}	0.1608

The calibrated HER and OER reference values are $\lambda_{H_2}^{\text{ref}} = 1.7347$ and $\lambda_{O_2}^{\text{ref}} = 1.5119$, respectively. These values define the thresholds below which hydrogen- and oxygen-

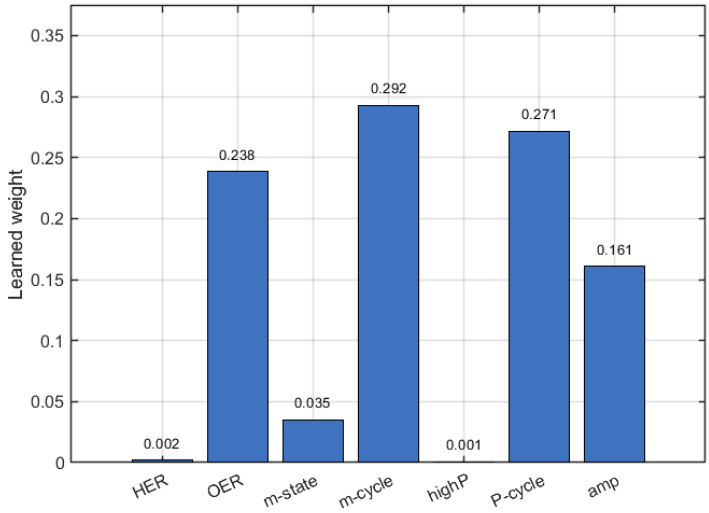


Figure 4.34: Learned weights of the extended degradation index.

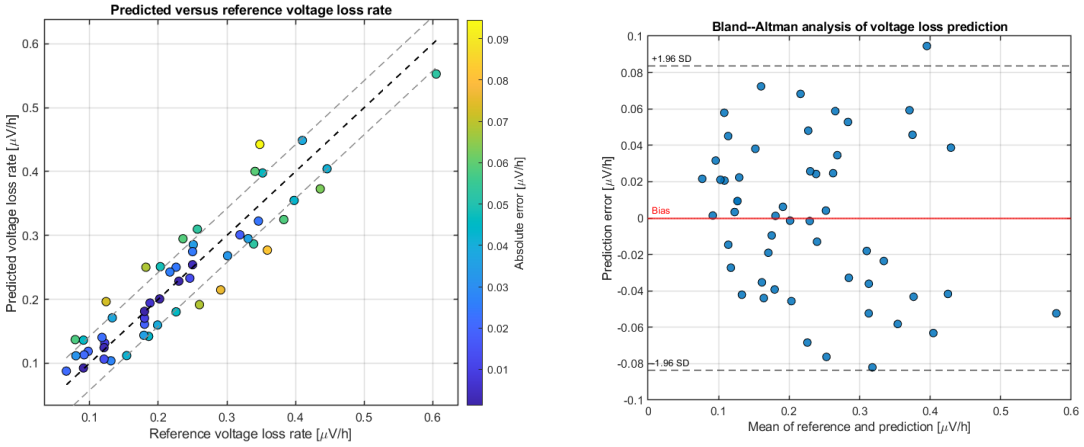
supply deficiencies contribute to the degradation index. The membrane water reference is identified as $\lambda_w^{\text{ref}} = 13.3165$, which represents the preferred hydration level around which membrane-water-content deviations are penalized. The calibrated high-power threshold is $P_{\text{high}}^{\text{ref}} = 180.0058$ kW, indicating that operation above this level is treated as high-load operation in the proposed severity formulation.

As shown in Table 4.16 and Fig. 4.34, the final degradation index is mainly influenced by membrane hydration cycling, power cycling, oxygen-supply-related severity, and power amplitude. The largest weight is assigned to the membrane hydration cycling term ($w_{m,\text{cycle}} = 0.2923$), followed by the power cycling term ($w_{P,\text{cycle}} = 0.2712$), the OER-related term ($w_O = 0.2384$), and the power-amplitude term ($w_{\text{amp}} = 0.1608$). This indicates that, within the investigated operating profiles, dynamic membrane-water variation, load transients, cathode-side oxygen-supply margin, and the operating range between low- and high-power levels are the most informative terms for reproducing the measured voltage loss rates.

In contrast, the HER term and the high-power threshold term receive very small weights, with $w_H = 0.0018$ and $w_{\text{highP}} = 0.0006$, respectively. This does not mean that hydrogen starvation or high-power operation is physically irrelevant. Rather, it shows that these two terms provide limited additional discriminative information for this particular data set once OER, membrane cycling, power cycling, and power amplitude are included. Similarly, the membrane state deviation term has a relatively small weight of $w_{m,\text{state}} = 0.0349$, indicating that the calibrated index is more sensitive to dynamic membrane hydration cycling than to the static deviation from the learned membrane-water reference.

4.3.2 Quantitative prediction accuracy

The quantitative prediction accuracy of the calibrated degradation index was evaluated by comparing the mapped degradation score with the experimental voltage loss rates. Fig. 4.35 shows the parity plot and the Bland–Altman plot of the final calibrated model, while Table 4.17 summarizes the main prediction accuracy metrics.



(a) Parity plot.

(b) Bland–Altman plot.

Figure 4.35: Quantitative prediction performance of the calibrated degradation index. The parity plot compares the predicted voltage loss rate with the experimental reference value, while the Bland–Altman plot shows the corresponding prediction error distribution.

Table 4.17: Quantitative prediction accuracy of the calibrated degradation index.

Metric	Value
Number of cases, N	53
Coefficient of determination, R^2	0.863
Root mean square error, RMSE	$0.042\ \mu\mathrm{V/h}$
Mean absolute error, MAE	$0.036\ \mu\mathrm{V/h}$
Pearson correlation coefficient	0.928
Spearman rank correlation coefficient	0.931

As shown in Fig. 4.35a, the predicted voltage loss rates generally follow the experimental reference values over the investigated degradation range. Most cases are distributed close to the ideal one-to-one line, indicating that the calibrated grey-box degradation index can reproduce the main variation trend of the experimental voltage loss rate. The dashed auxiliary lines indicate the RMSE band, which provides a visual reference for the typical prediction deviation.

The numerical results in Table 4.17 further confirm this agreement. The calibrated model achieves an R^2 value of 0.863, with an RMSE of $0.042\ \mu\mathrm{V/h}$ and an MAE of $0.036\ \mu\mathrm{V/h}$. These values show that the mapped degradation index provides a

reasonable quantitative approximation of the measured voltage loss rate within the investigated operating profiles. In addition, the Pearson correlation coefficient reaches 0.928, indicating a strong linear agreement between prediction and reference. The Spearman rank correlation coefficient is 0.931, which shows that the model also preserves the relative degradation ranking among different operating cases.

The Bland–Altman plot in Fig. 4.35b provides a complementary view of the prediction errors. The mean bias is approximately zero, suggesting that the final mapping does not introduce a systematic overestimation or underestimation of the voltage loss rate. Most samples lie within the agreement limits, and the remaining deviations are mainly case-dependent scatter rather than a consistent directional error.

Overall, these results indicate that the calibrated degradation index is able to map the proposed internal-state and power-history severity terms to the experimentally observed voltage loss rate with good numerical accuracy and strong ranking consistency. Therefore, the index is suitable for comparing the relative degradation severity of different operating profiles and for serving as a health-related cost term in the subsequent analysis.

4.3.3 Comparison with empirical baseline

To further assess the advantage of data-driven calibration over empirically assigned coefficients, an empirical baseline model is constructed following the grey-box perspective introduced in [3]. The baseline retains the same four internal-state severity terms as the proposed framework, namely the hydrogen starvation term J_H, the oxygen starvation term J_O, the membrane water-content state deviation term $J_{m,\text{state}}$, and the membrane hydration cycling term $J_{m,\text{cycle}}$, but assigns equal weights of 0.25 to each term rather than calibrated weights. Since the original work in [3] is control-oriented and does not explicitly report quantitative degradation weights, equal weighting represents the most natural approximation of an empirical coefficient assignment. Both models apply the same min–max normalisation across the 53 calibration profiles and the same quadratic score-to-voltage-loss-rate mapping to ensure a fair comparison.

The quantitative results are summarised in Table 4.18, and the signed prediction error distributions are compared in Fig. 4.36.

Table 4.18: Quantitative comparison between the empirical baseline and the proposed calibrated degradation index.

Metric	Empirical baseline	Proposed model
RMSE [μV/h]	0.071	0.042
MAE [μV/h]	0.059	0.036
R^2	0.612	0.863
Pearson correlation	0.780	0.928
Spearman correlation	0.716	0.931
Pairwise ranking acc.	78.0%	91.5%
Top-10 hit rate	60.0%	90.0%

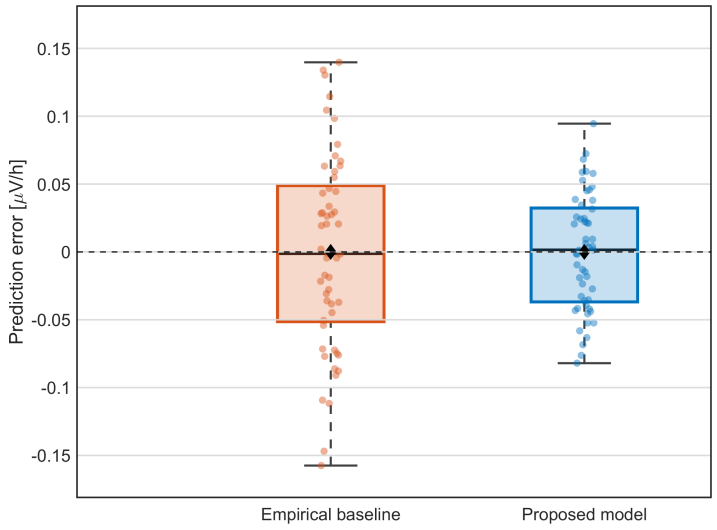


Figure 4.36: Signed prediction error distributions of the empirical baseline and the proposed calibrated model across the 53 degradation profiles. The diamond marker indicates the mean error.

The proposed calibrated model outperforms the empirical baseline across all reported metrics. The RMSE is reduced from 0.071 μV/h to 0.042 μV/h, a reduction of approximately 40%, and the coefficient of determination improves from 0.612 to 0.863. The Spearman rank correlation coefficient increases from 0.716 to 0.931, indicating substantially better preservation of the relative degradation ranking across operating profiles. The pairwise ranking accuracy improves from 78.0% to 91.5%, and the top-10 hit rate increases from 60.0% to 90.0%, further confirming the superior ranking consistency of the proposed model.

As shown in Fig. 4.36, both models exhibit a near-zero median prediction error, indicating that neither introduces a systematic bias. However, the interquartile range of the proposed model is substantially narrower than that of the empirical baseline, reflecting more consistent prediction accuracy across different operating profiles. The wider error spread of the empirical baseline is attributed to the inability of equal weights to capture the differing contributions of each severity term to the voltage loss rate under varying dynamic operating conditions.

These results demonstrate that the data-driven calibration procedure proposed in this work, which simultaneously optimises the severity thresholds and term weights against experimentally measured voltage loss rates, provides a more accurate and consistent degradation assessment than an approach relying on empirically assigned equal weights. This directly addresses the second research question by showing that calibrated coefficients can better quantify the individual contributions of the degradation-related severity terms to the stack voltage loss rate.

Chapter 5

Conclusions and Future Work

5.1 Conclusions

This thesis developed a simulation-informed grey-box degradation modelling framework for proton exchange membrane fuel cell systems. The main objective is to estimate degradation-relevant operating conditions without large computational cost or direct internal-state measurements, and to map the directly to voltage loss rate. To achieve this, a calibrated Simscape PEMFC system model was first used to generate physically meaningful internal state variables. Machine-learning models are then trained to estimate hydrogen excess ratio (HER), oxygen excess ratio (OER), and membrane water content from fuel-cell power-demand features. Finally, these estimated internal states were combined with power-profile severity terms to construct a grey-box degradation index, which is calibrated against experimentally measured voltage loss rates.

The first research question addressed how the choice of regression model and input feature configuration affects the estimation accuracy of HER, OER, and membrane water content under dynamic load profiles. The results show that random forest regression provided the most reliable external generalization among the evaluated models. Although SVR achieved competitive performance in some segment-level cross-validation cases, RF gave the strongest overall performance on the independent validation set and was therefore selected as the representative model for the final internal-state estimator.

For HER prediction, the compact feature set consisting of the current fuel-cell power and power slope, $P_k + \dot{P}_k$, was sufficient to obtain good external generalization. The RF model achieved an external R^2 value of approximately 0.905 using this compact feature set. Although history-dependent features further improved the numerical external accuracy, they also introduced local non-smooth artefacts in some transient regions. Therefore, $P_k + \dot{P}_k$ was selected as the preferred HER input representation because it provides a better balance between accuracy, simplicity, physical interpretability, and real-time feasibility.

For OER prediction, the same compact feature set, $P_k + \dot{P}_k$, performed even more strongly. The RF model achieved an external R^2 value of approximately 0.996, and the additional history-dependent features provided only marginal improvement. This indicates that the oxygen excess ratio is mainly governed by the current operating point and the instantaneous load-changing trend within the investigated operating range. Therefore, P_k+

$\dot{P}_k$ was also selected as the preferred input configuration for OER prediction.

In contrast, membrane water content showed a stronger dependence on recent operating history. The compact instantaneous feature sets failed to generalize reliably to the external profiles, whereas the 40 s history-based feature set substantially improved the prediction performance and achieved an external R^2 value of approximately 0.901. This result is physically consistent with the slower dynamics of membrane hydration, which are affected by water generation, removal, diffusion, humidification, and previous load conditions. Therefore, the final feature configuration uses $P_k + \dot{P}_k$ for HER and OER, and $P_k + \dot{P}_k$ together with 40 s history-dependent descriptors for λ_w.

The second research question addressed the relative contribution of different degradation-related severity terms to the experimentally measured voltage loss rate. The calibrated degradation index was able to reproduce the main variation trend of the experimental voltage loss data. The final model achieved an R^2 value of 0.863, an RMSE of 0.042 μV/h, an MAE of 0.036 μV/h, a Pearson correlation coefficient of 0.928, and a Spearman rank correlation coefficient of 0.931. These results show that the proposed index provides not only a reasonable quantitative approximation of the measured voltage loss rate, but also strong ranking consistency among different operating profiles.

The calibrated weights further indicate that the most influential degradation-related terms are membrane hydration cycling, power cycling, oxygen-supply-related severity, and power amplitude. The largest contribution was assigned to membrane hydration cycling, followed by power cycling, OER-related severity, and power amplitude. This suggests that, within the investigated operating profiles, dynamic water-management variation and load transients are the dominant degradation-driving factors, while cathode-side oxygen-supply margin and the overall power cycling range also provide important discriminative information.The high-power threshold term likely provided limited additional information because high-load effects were already partly represented by power cycling and power amplitude. In contrast, the small HER-related weight is more likely caused by the limited coverage of hydrogen-starvation conditions in the calibration profiles. Therefore, the result does not imply that hydrogen starvation is physically irrelevant, but only that HER-related severity was not a dominant discriminating factor within the available degradation dataset.

Overall, the proposed framework provides a physically interpretable and data-efficient approach for PEMFC degradation assessment. Compared with purely data-driven degradation prediction, the framework preserves an explicit connection to internal operating states and degradation-prone mechanisms. Compared with detailed physics-based ageing models, it does not require complete knowledge of internal stack parameters or material-level degradation laws. It introduced the effect of historical power features into the prediction of HER, OER and membrane water content. Additionally, the grey-box degradation index in this work links measurable dynamic power-demand information directly to observed voltage loss rates through physically meaningful internal-state and load-severity descriptors.

5.2 Limitations

Howerver, several limitations should also be acknowledged. First, the machine-learning estimators for HER, OER, and membrane water content were trained using data generated from the calibrated Simscape model rather than from direct measurements of these internal variables. It means that the learned estimators inherit the assumptions and possible modelling errors of the simulation model. In particular, prediction accuracy may be limited in operating regions where the calibrated Simscape model shows residual mismatch, such as parts of the low-power region.

Additionally, the degradation-index calibration was based on only 53 operating cases. These cases cover only simple types of dynamic cycling profiles. More complex operating conditions may induce richer internal-state variations, such as hydrogen and oxygen supply transients, membrane hydration fluctuations, and temperature-dependent effects; therefore, the calibrated weights may not fully represent the degradation behaviour under broader real-world operating conditions.

Moreover, the proposed framework was evaluated under a specific PEMFC system model, a specific reference efficiency curve, and a specific set of degradation calibration profiles. Its transferability to other fuel-cell stacks, different system architectures, different control strategies, or different environmental conditions has not yet been fully assessed.

5.3 Future Work

Future work should first focus on experimental validation of the internal-state estimators. If HER, OER and membrane water content can be obtained from a real PEMFC test bench, the machine-learning predictions can be independently evaluated against measured data. This would also allow the error propagation from internal-state estimation to the final degradation index to be quantified.

Second, the degradation calibration dataset should be extended to cover more diverse and more severe operating conditions. In particular, profiles involving hydrogen starvation, strong oxygen-supply transients, wider power-amplitude ranges, and different thermal or humidity conditions should be included. This would help verify whether the calibrated weights remain valid under broader operating conditions, especially for the HER-related term, which received a very small weight in the current dataset.

Finally, the proposed degradation index could be integrated into an energy management or model predictive control framework as a health-related cost term. This would make it possible to evaluate whether penalizing degradation-prone operation, can reduce long-term fuel-cell degradation.

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